

The supernova history of Cygnus OB2 from *Gaia* DR3

A branch-resolved census of massive-star deaths

V. Voitsekховskiy¹ and (co-authors to be confirmed)¹

Affiliation to be completed e-mail: jstvd@gmail.com

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ABSTRACT

Context. The supernova history of a young association sets the mechanical energy driving its superbubble, its compact-remnant population, its radioactive-isotope budget, and the viability of any interpretation of its high-energy emission that invokes a recent explosion. It is hard to measure because it couples membership, extinction, age, completeness and high-mass stellar evolution, and existing numbers for Cygnus OB2 are population-synthesis estimates built on heterogeneous literature inputs.

Aims. We reconstruct the massive-star and supernova history of Cygnus OB2 from a single homogeneous census, and separate the conclusions the data control from those the carried model choices control.

Methods. From *Gaia* DR3 astrometry and photometry joined to 2MASS and to a literature spectroscopic census we build a probabilistic member list, decompose it kinematically, fit per-star extinctions against a spectroscopically anchored kriged spatial prior, infer subgroup ages and per-star masses from two isochrone families, and normalize the initial mass function by a response-aware Poisson fit to star counts between 2 and 8 $M_{\odot}$. The normalization is tested out of sample against the observed census above 8 $M_{\odot}$. Supernovae are counted by stellar-evolution turnoff in a Monte Carlo over 2×10^6 population realizations per branch. Isochrone family, extinction law, IMF slope, star-formation duration and explodability are carried as explicit branches and never averaged.

Results. Cygnus OB2 resolves into three stable kinematic subgroups of 4.00, 4.09 and 2.52 Myr, two older and one younger, with a systematic age envelope far wider than those three central values. Conditional on the permissive all-explode prescription, the baseline gives $N_{\text{SN}} = 8.43$ (median 8, 68% [5, 11]) and the 36 headline branches span 5.63–28.7; the model-branch spread exceeds the stochastic interval of any one branch. The corresponding probability that the most recent event falls within 100 kyr is 0.552 on the baseline and spans 0.411–0.889. Every inferred stellar death lies above 33.9 $M_{\odot}$, so the ledger measures very-massive-star explodability rather than the canonical regime. Association-wide closure is near Salpeter, but the subgroups are heterogeneous: their closing slopes span 2.06–2.34.

Conclusions. The stellar census alone does not decide whether these very massive deaths made supernovae: a hard direct-collapse cutoff at or below 30 $M_{\odot}$ returns zero. PSR J2032+4127 independently excludes an entirely explosion-free history if its association with Cygnus OB2 is accepted. The combined evidence therefore requires either a successful explosion in the very-massive regime or an older or binary-stripped progenitor channel absent from the coeval single-star ledger. The supernova-driven interpretation of the Cygnus γ -ray cocoon is evaluated as a conditional scenario-availability score rather than a calibrated probability; it is controlled by the high-mass IMF slope and by two terms that are a model mapping and a bound rather than measurements. *Gaia* DR4 will sharpen ages and runaways but not the high-mass IMF; a COSI measurement or upper limit would constrain the joint IMF and very-massive-star yield problem.

Key words. stars: massive – open clusters and associations: individual: Cygnus OB2 – supernovae: general – gamma rays: ISM – astrometry

1. Introduction

1.1. Why young-association supernova histories matter

How many stars an association has already lost, and when, is a population property that several distinct fields depend on. It sets the mechanical energy injected into the surrounding superbubble and therefore the structure of the cavity; it sets the expected number of compact remnants; it sets the supernova contribution to the radioactive ^{26}Al and ^{60}Fe budgets (Knödlseeder et al. 2002; Martin et al. 2010); and it decides whether interpretations of high-energy emission that require a recent explosion are admissible at all.

That last dependence has become acute for Cygnus OB2. The γ -ray cocoon around it, detected from GeV to hundreds of TeV, is the clearest Galactic PeVatron candidate, and the proposed accelerators span the association's collective winds,

a supernova exploding into the cavity those winds have carved (Härer et al. 2025; Li et al. 2026), and — since LHAASO's detection of a variable ultra-high-energy source coincident with Cygnus X-3 — a microquasar (Shi et al. 2026). The field also hosts the γ Cygni supernova remnant and the pulsar PSR J2032+4127 with its wind nebula. These candidates are not mutually exclusive, but they place different demands on the stellar population, and the supernova-driven class places the sharpest one: the fiducial model of Härer et al. (2025) uses a powerful Type Ic event in the cavity at roughly 50 kyr, with an energy of a few 10^{51} erg. Those are model-dependent requirements rather than direct observables: their permitted age and energy change with the assumed cavity density, acceleration efficiency and particle cut-off.

1.2. Why the measurement is hard

An association’s present-day census is not its initial population, and the mapping between them runs through every step of the analysis. Extinction in Cygnus is severe and differential, so a single regional value biases the inferred masses. Membership must be probabilistic, because a bright O star with poor astrometry cannot be allowed to fall out silently. Completeness is a function of both mass and sky position, so a scalar correction is not the right object. The sampling of the upper mass function is stochastic at these numbers. And the two quantities that convert a census into a death toll — the age and the fate of the most massive stars — are precisely the two the data constrain least well.

1.3. Why Cygnus OB2

Cygnus OB2 is one of the most massive young associations in the Galaxy, with tens of O stars, three Wolf–Rayet stars and a young pulsar (Wright et al. 2015; Berlanas et al. 2020). It has a rich enough massive-star population for the mass function to be measured rather than assumed, severe enough differential extinction to make the extinction treatment load-bearing, substructure that lets the assumption of coevality be tested rather than imposed, and independent supernova markers that can be compared against a ledger built without reference to them.

1.4. What previous estimates could and could not do

Prior estimates disagree in ways that trace to their inputs rather than to their methods. Knödseder et al. (2002) and Martin et al. (2010) modelled the Cygnus complex as a whole from heterogeneous literature counts and assumed ages; Menchiari et al. (2024) adopt a fixed association mass and predict 7 ± 2.5 supernovae at 3 Myr and 26 ± 5 at 5 Myr — a factor of four spanned by a 2 Myr change in an assumed age. The lesson we take from this is not that those numbers are wrong but that a supernova count is only interpretable together with the machinery that says what it is sensitive to.

1.5. What this paper adds

The stellar side of the problem is tractable. If the present-day mass function is measured where it is complete, the initial mass function can be normalized there and extrapolated above the turnoff, and the stars missing from the upper main sequence are the ones that have died. This is the deficit method of Fuchs et al. (2006) and Zucker et al. (2022), and *Gaia* DR3 makes it applicable to Cygnus OB2 with membership, distance and extinction all derived from the same homogeneous data set.

The methodological contribution is that the whole chain is internally linked, branch-resolved and provenance-controlled: the isochrones that set the extinctions are the isochrones that set the ages; the completeness response is measured by injection through the actual selection rather than assumed; the normalization is tested out of sample; and every discrete modelling choice is carried as a branch to the end rather than marginalized into a single number. Figure 1 shows the chain, marks which nodes are observations and which are inferences, and marks where a failed validation test or a one-sided bound enters.

The cocoon and radioactive-isotope questions then appear as applications of that ledger — demonstrations of what it can and cannot decide — rather than as the premise of the work. Three choices govern everything and are stated here rather than buried: discrete modelling choices are *carried as branches and never*

averaged; every comparison against an independent supernova marker was *pre-registered* before it was computed; and results that failed are reported as failed.

This conditional structure is the scientific result rather than an absence of one. The census identifies which stars could already have died, the branch grid quantifies the supernova history if those deaths explode, and the pulsar then excludes the all-zero outcome under the association hypothesis. What remains is a physical alternative: either at least one star in the very-massive regime exploded successfully, or the pulsar came through an older or binary-modified channel not represented by the coeval single-star calculation. The analysis therefore converts a poorly localized uncertainty in “the supernova number” into a testable constraint on high-mass evolution.

2. Data

2.1. Gaia DR3 and 2MASS

Two *Gaia* DR3 queries define the working samples: a narrow box ($l \in [77^\circ, 83^\circ]$, $b \in [-1^\circ 5', +4^\circ]$) for the census, and a wide box ($l \in [72^\circ, 88^\circ]$, $b \in [-5^\circ, +8^\circ]$) for run-away traceback. Parallaxes were selected in $[0.35, 1.1]$ mas and $G < 19$ — deliberately generous, because the selection is refined by clustering rather than by cuts. Parallax zero-points follow Lindegren et al. (2021). Near-infrared photometry comes from the official *Gaia* $\times$ 2MASS cross-match. Table A.1 summarizes the resulting subgroups.

2.2. Spectroscopic anchors

The brightest O stars have unreliable or missing *Gaia* astrometry, so they cannot be allowed to fall out of the census silently. A merged anchor table carries spectral types and, where available, effective temperatures from Wright et al. (2015), Berlanas et al. (2019, 2020) and the Galactic O-Star Catalog. Source identities and literature-evidence rows are distinguished throughout: the anchor table resolves to fewer unique *Gaia* sources than it has literature rows, because several stars are catalogued more than once. The spectral reclassification of the association’s B0 population by Galán-Diéguez et al. (2025) postdates our freeze and should be folded in at the DR4 rerun.

Anchors are exempt from the astrometric quality cuts and are treated individually. That exemption is a deliberate methodological choice with a measurable cost, and Sect. 4.1 reports its size rather than absorbing it.

2.3. Independent supernova markers

PSR J2032+4127 from the ATNF catalogue, γ Cygni (G78.2+2.1) from Green’s supernova-remnant catalogue, and the INTEGRAL ^{26}Al measurements of the Cygnus region were frozen at the data-acquisition stage, *before* any ledger existed, and their comparison statistics were fixed in a pre-registration written before the ledger was run. The ^{60}Fe forecast of Sect. 5.8 is *not* in this category: it was chosen after the ledger existed. The two are kept apart everywhere they are used.

3. Method

3.1. Membership selection

Association membership is decided by a two-component cluster-versus-field Gaussian mixture in five features: Galactic position

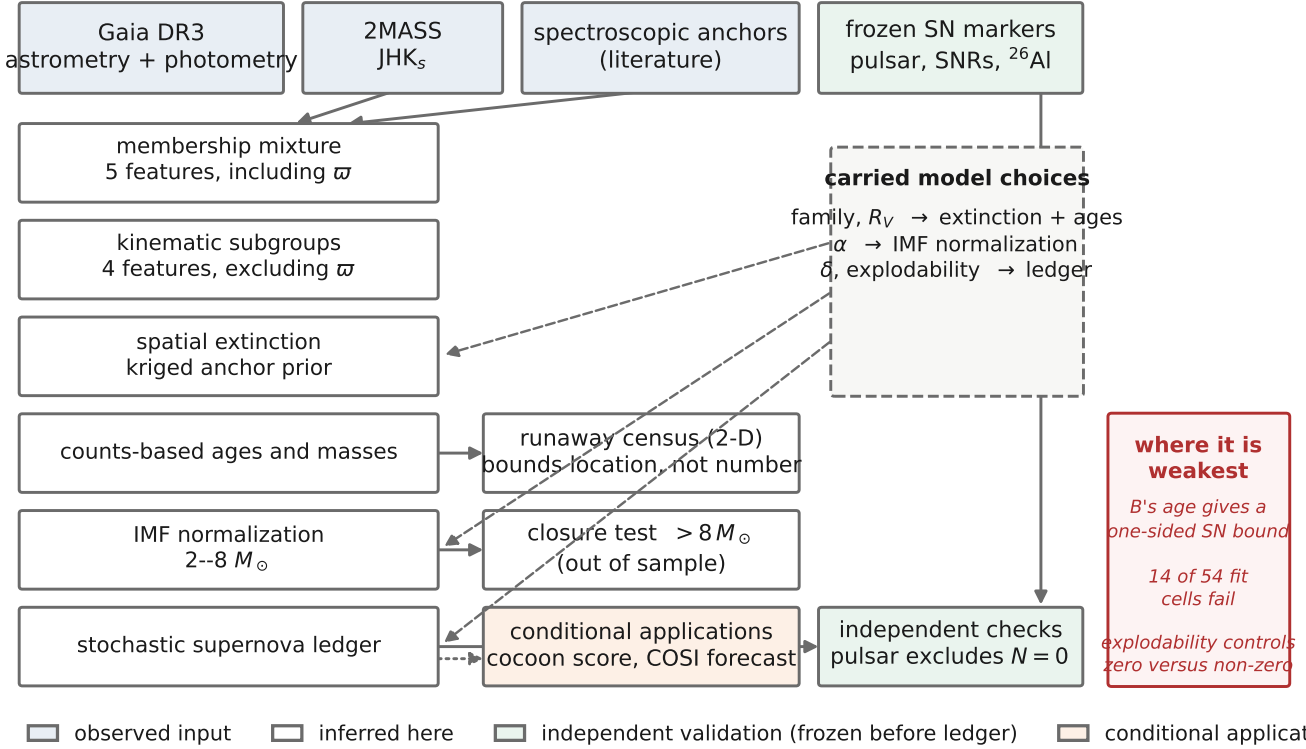


Fig. 1. The inference chain and its evidence hierarchy. Blue nodes are observed inputs, white nodes are quantities inferred here, green nodes are independent validation frozen at the data-acquisition stage before any ledger existed, and the orange node holds the conditional applications. The connectors from the model-choice panel show the stage at which each branch axis enters the chain. The red panel names the three places where the inference is weakest and where the text is correspondingly qualified.

relative to the field centre, the zero-point-corrected parallax, and the two proper-motion components. A DBSCAN pass over the same space provides the training seed only; it assigns no final label. Membership probabilities come from a Monte Carlo over each star's full *Gaia* astrometric covariance, so a star with large but consistent errors is treated as uncertain rather than as excluded. The mixture prior was calibrated against three control fields at matched Galactic latitude processed through the identical pipeline. The mixture publishes a continuous probability, but a finite catalogue is needed by the photometric and age pipeline. We use $P > 0.5$ for that handoff: for a two-class posterior it is the maximum-posterior decision rule, equivalent to requiring that membership be favoured over the field under equal misclassification costs. It is an operational choice, not a universal astrophysical threshold. Inside the selected catalogue the continuous probability is retained and used wherever the downstream estimator calls for a membership weight; spectroscopic anchors that fail the astrometric quality cuts enter as explicitly flagged manual exceptions.

The threshold is checked rather than justified by convention. Moving it from 0.4 to 0.6 changes the clean automatic catalogue from 1,441–1,186 stars. Across that interval the automatic recovery of the external membership list changes only from 77.4% to 75.6%, while the mean control-to-target yield falls from 5.4% to 2.8%. The adopted 0.5 boundary retains 1,331 automatic members, with 76.2% automatic recovery and a 3.7% mean control yield (Appendix B). A stricter cut buys purity at the cost of a smaller calibration sample; a looser cut does the reverse.

Two limitations follow directly and are carried to the end. First, because the control fields calibrated the prior, their measured yield is not a wholly external false-positive rate. Second, because parallax is a selection feature, any test for structure along the line of sight is conditional on the sample the selection produced (Sect. 3.3).

3.2. Kinematic subgroups

Subgroup decomposition is a *separate* operation on the clean automatic member sample, in a *four*-dimensional feature space ($l, b, \mu_{\alpha*}, \mu_{\delta}$) from which parallax is deliberately excluded. The exclusion belongs here and not at the membership stage: the members form a single population at 1.62 kpc with an intrinsic depth of order 45 pc, which DR3 parallaxes cannot resolve. Admitting parallax as a clustering feature could therefore make the recovered labels primarily reflect measurement noise and the preceding selection rather than physical depth.

The number of components was accepted on seed stability rather than on an information criterion; only $k = 3$ is stable across 50 deterministic seeds, and per-star label confusion over those seeds is below a fraction of a per cent. HDBSCAN was scanned for completeness but not used for labels. Anchors that carry no astrometric label remain unassigned and are carried for provenance.

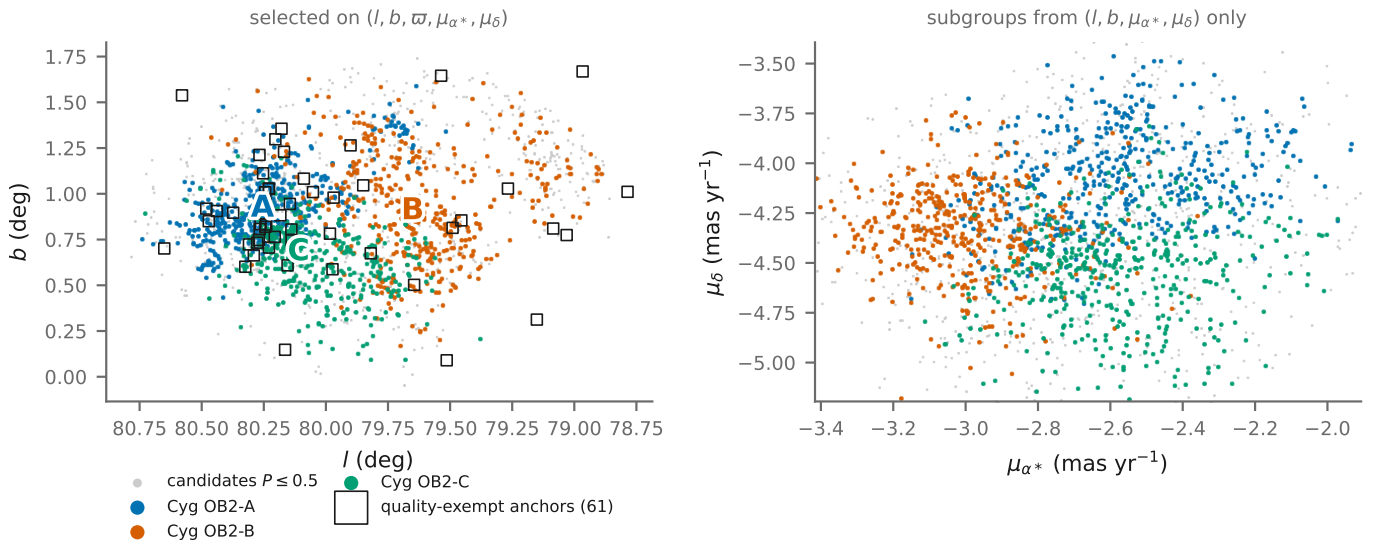


Fig. 2. Membership and substructure, showing the two operations separately. *Left:* sky distribution of the 1,392 members with $P > 0.5$, coloured and directly labelled by kinematic subgroup, with lower-probability candidates in grey and the manual quality-exempt spectroscopic anchors marked by open squares. Association membership used $(l, b, \varpi, \mu_{\alpha*}, \mu_{\delta})$, including parallax. *Right:* the proper-motion plane. The subgroup labels themselves come from $(l, b, \mu_{\alpha*}, \mu_{\delta})$ only, with parallax excluded at this stage and not at the previous one.

3.3. The distance test and its exact scope

Within the selected member sample we compare a one-component and a two-component distance model and find no support for the second. We state the limit of that non-confirmation precisely: parallax was a selection feature, so a genuine foreground group would have been removed before the test ran. The defensible claim is that *there is no evidence for two distances among the members we select*, not that the foreground population of [Berlanas et al. \(2019\)](#) does not exist. A parallax-blind membership rerun is the only way to settle it; it is deferred because it requires re-deriving the entire downstream chain on a different member sample, which is a second paper's worth of work rather than a robustness check.

3.4. Extinction

Differential extinction is the Cygnus problem, and a single regional A_V is inadmissible. Anchors receive A_V from fixed- T_{eff} intrinsic colours, which avoids the temperature-extinction degeneracy that broadband photometry suffers for hot stars. All other members are fitted against reddened synthetic photometry from the same isochrone families used for the ages, in six bands with missing bands masked rather than imputed, under an anchor-informed spatial prior.

That prior's mean is a simple-kriging estimate over the eight nearest anchors using a variogram fitted to the anchors themselves, and its width is the unconditional variogram sigma. This replaced an earlier construction whose two moments contradicted each other: it widened the prior for stars far from any anchor, correctly declaring those anchors uninformative, while still centring on them at full weight. The repair matters for one subgroup only, and for an observational reason rather than a modelling one — Cyg OB2-B's local anchor support is far sparser than its siblings' (Fig. 3).

One systematic is carried rather than removed: at matched sky position, prior-free broadband photometry prefers an A_V about half a magnitude below the spectroscopically calibrated

anchors. The absolute extinction scale, and hence the mass scale and N_{SN} , therefore rests on the anchor calibration.

3.5. Ages and masses

PARSEC and MIST are carried in parallel from the start, never merged. Two age indicators are used, an upper-main-sequence fit and a counts-based fit, and both are retained. Of the age branches attempted a minority are excluded outright or rail against the grid edge, and those are reported rather than dropped (Appendix C). Isochrones are interpolated between native table ages in log space rather than snapped to the nearest tabulated age, because a stepped turnoff would make N_{SN} a step function of age.

The upper-main-sequence ages (3.98, 3.55, 2.51 Myr) are the prior, not the adopted values. What the ledger consumes is the counts-based age of Sect. 3.6, which is inferred jointly with the normalization because the two are correlated through the fit. Only Cyg OB2-B moves materially between the two definitions, by +0.54 Myr, and that shift is worth a factor 1.92 in B's supernova count — which is why both are reported and labelled (Fig. 5).

B's counts-based posterior concentrates 53% of its weight on the topmost of its nine prior nodes. A posterior that piles up against the edge of its prior grid reports a bound, not a measurement, and the direction is known: older means a lower turnoff means *more* supernovae. B's contribution to N_{SN} is therefore a one-sided lower bound, and Cyg OB2-C sits on a family-dependent boundary at which its turnoff crosses the IMF's upper limit.

3.6. Completeness and IMF normalization

The normalization k is fitted per subgroup by a Poisson likelihood over mass bins between 2 and 8 $M_{\odot}$, against a forward model that convolves the assumed IMF with a completeness response measured by injecting synthetic stars into the real field and recovering them through the actual selection, end to end at catalogue level. The response, not a scalar completeness, is

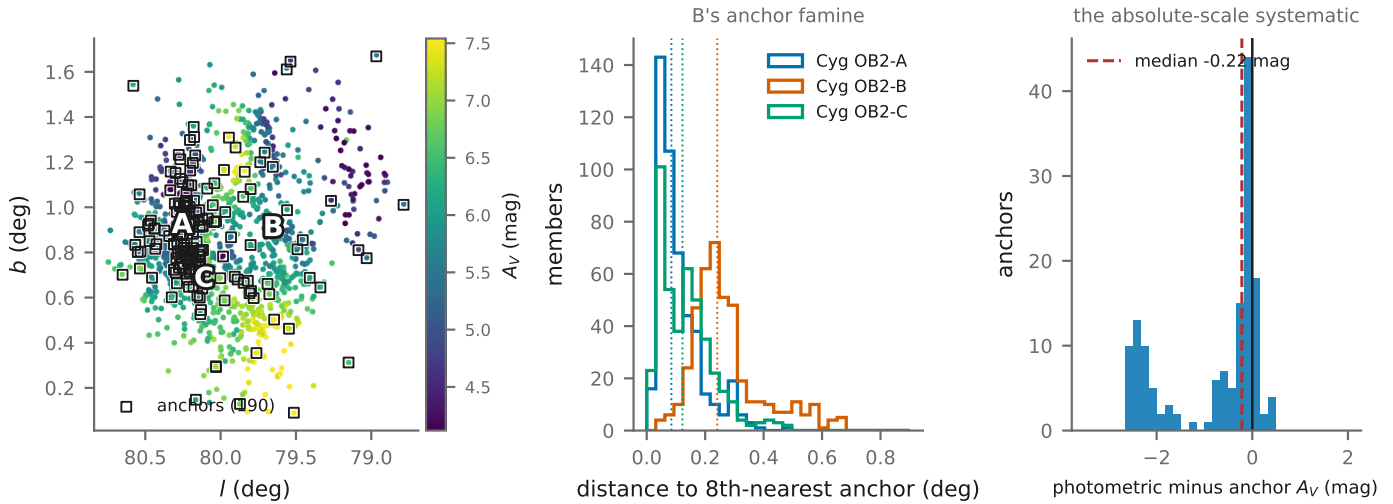


Fig. 3. Extinction and the support behind it. *Left:* the per-star A_V map on the baseline branch, with spectroscopic anchors as open squares and the three subgroup centroids labelled. *Centre:* distance to the eighth-nearest anchor, per member, by subgroup; Cyg OB2-B’s distribution is displaced to markedly poorer local calibration, which is why the kriged prior changed B and left A and C essentially untouched. *Right:* the same-star difference between the prior-free photometric A_V and the anchor A_V , which is the absolute-scale systematic the mass scale inherits.

the correct object: the mass estimator has 22–25% scatter, so it moves stars across bin edges and across both window boundaries. No subgroup reaches an absolute 95% completeness edge, which is why a completeness cut was never an option.

The truth-side age is marginalized jointly with k over nine age nodes drawn from the CMD posterior and reweighted by the Poisson likelihood of the observed counts with k integrated out analytically. The binary fraction is mass-dependent in the final version of the injection chain.

A branch passes the residual-fit criterion only if its χ^2 probability, its residual trend probability and its worst absolute Pearson residual are all inside their thresholds, in all three subgroups. Sect. 4.3 reports what that validation actually returns, which is not what a reader would infer from the baseline alone. Failed branches are retained as explicit sensitivity cases and never deleted, because deleting a branch because it fits badly under one assumption is a way of selecting the assumption.

3.7. Massive-star closure

The closure test compares the number of living stars above $8 M_\odot$ predicted by the normalization against the number observed. It is *out of sample*: k is fitted from 2– $8 M_\odot$ counts and the $> 8 M_\odot$ census never enters that likelihood. Both sides are computed with the same forward response, integrated with no lower cut-off, because the observed side counts every member’s $P(M > 8 M_\odot)$ whatever its true mass; truncating that integral at the threshold the response smears across was a real defect in an earlier version and is recorded as such (Appendix G). The observed side combines members, orphan anchors inside the footprint but outside the member sample, and false-positive-corrected runaways.

Following the convention of the underlying work, the closure ratio is *observed over predicted*, so a value above unity means the census contains more massive stars than the extrapolated normalization predicts.

3.8. Runaways

Runaways are recovered from the wide box by tracing *peculiar* proper motions backwards over the subgroup age windows, with the association’s systemic motion subtracted — Cygnus OB2’s bulk motion is larger than a typical ejection signature, so absolute proper motions measure drift rather than ejection. The false-positive rate is measured per separation bin from control fields, in the same spirit as recent *Gaia* runaway searches in other associations (Azatyan et al. 2026).

The recovered count is a lower bound twice over: the traceback is two-dimensional, so line-of-sight ejections are missed, and the finite wide box caps recovery at a velocity–time ceiling. Runaways do *not* enter N_{SN} . They are living stars below the turnoff, so they change neither k , which is fitted below $8 M_\odot$, nor the turnoff mass. What they bound is location, not number — and applying a fraction measured on living, lower-mass stars to dead, more massive progenitors is an assumption, not a measurement (Sect. 5.4).

3.9. The supernova ledger

For each branch and each of 2×10^6 iterations we draw a paired (k , age) sample from the joint posterior — paired, because the two are correlated through the fit — draw a star-formation duration, sample a discrete stellar population from the IMF conditioned on the calibration-window counts, and ask which stars have passed the turnoff. The stellar lifetime is obtained by inverting the project’s own turnoff relation, so the ledger and the ages cannot disagree about when a star of a given mass dies. A death becomes a supernova only if it satisfies the explodability branch.

Two explodability prescriptions are carried. The first is *all explode*. The second is a *hard direct-collapse cutoff*, in which no star above a fixed mass produces a supernova; the scan over that threshold, not the single adopted value, is the primary explodability result (Sect. 4.6). We emphasise that this is a hard cutoff and not an implementation of the non-monotonic explodability maps of Sukhbold et al. (2016) and Ertl et al. (2016); no detailed high-mass explodability prescription was implemented here.

3.10. Branches, and what is never averaged

The branch grid carries five modelling decisions, each for a different physical reason.

Stellar-evolution family: PARSEC and MIST are carried separately because their high-mass turnoffs and colour–magnitude relations are independent model calibrations. Their difference is a model systematic, not a probability distribution to average.

Extinction law: $R_V \in \{3.0, 3.1, 3.5\}$ spans the canonical diffuse-ISM value and the plausible range across the Cygnus sightline. The same value is used consistently when deriving extinction, age, mass and completeness.

IMF slope: $\alpha \in \{2.0, 2.3, 2.6\}$ brackets a shallow high-mass population, a Salpeter-like reference, and a steep sensitivity case. This is the axis to which both the massive-star closure and the supernova count are most sensitive.

Formation duration: $\delta \in \{0, 1, 2\}$ Myr compares a coeval population with modest internal age spreads. A wider duration is not carried because it would cease to be a perturbation around the independently inferred subgroup ages.

Explodability: the permissive all-explode prescription counts every stellar death as a supernova. The hard direct-collapse scan instead asks how the answer changes if stars above a threshold fail. The latter is a diagnostic family of cutoffs, not a probability-weighted physical explodability model.

The baseline branch is PARSEC, $R_V = 3.1$, $\alpha = 2.3$, $\delta = 0$, all explode. It is a transparent reference coordinate: the canonical extinction law, Salpeter-like slope, minimal coeval assumption and maximally permissive explodability prescription. It is not asserted to be more probable than the other branches. The full tabulated grid contains $2 \times 3 \times 3 \times 3 = 54$ all-explode branches; each can then be re-evaluated against the direct-collapse threshold scan. No probability weights exist for these discrete choices, so they are never averaged into a pseudo-posterior.

The headline all-explode branch set excludes $\alpha = 2.6$. The criterion is the calibration-window χ^2 , an internal fit statistic fixed long before the ledger existed, on which $\alpha = 2.6$ is the best fit in 1 of 18 cells and has the worst median ($\chi^2 = 10.80$ against 6.86 for $\alpha = 2.3$). The census closure agrees independently but was deliberately *not* used to make this choice, so that it remains available as an out-of-sample validation of the extrapolation (Sect. 4.4). $\alpha = 2.6$ is reported throughout with its own numbers ($N_{\text{SN}} = 1.93\text{--}4.24$) and is never deleted.

The headline set is selected on α and on nothing else. In particular it is *not* selected on mass-function fit quality, and Sect. 4.3 reports how many of its branches rest on fits that passed their own fit-quality criterion.

4. Results

4.1. Membership, contamination and kinematic substructure

Of 2,112 catalogue rows, 1,392 have $P > 0.5$ and 1,331 carry a subgroup label: Cyg OB2-A (476), Cyg OB2-B (426) and Cyg OB2-C (429). The three subgroups are stable in the sense defined in Sect. 3.2 and are recovered with the same membership at every seed.

The recall benchmark against Berlanas et al. (2019) needs stating in two parts rather than one. Of their 168 published members that fall inside our quality-analyzable sample, the automatic full-covariance posterior recovers 128, or 76.2%. Adding the 61 manual spectroscopic quality exceptions brings total recall to

189 of 229, or 82.5%. Manual treatment of the anchors was required by the design and is not a protocol violation, but the total is therefore *not* an independent machine-classifier recall, and we do not present it as one. Appendix B gives the accounting.

The control fields yield 3.7% of the association field’s members at $P > 0.5$ after area normalization (Fig. 4). Because those same control fields calibrated the mixture prior, this is a strong internal consistency check but not a fully external false-positive rate.

Within the selected sample we find no evidence for two distinct distances (Sect. 3.3), with the scope of that statement as given there.

4.2. Differential extinction and subgroup age structure

Extinction is spatially coherent and cannot be represented by a single regional value; the median A_V differs between subgroups by more than a magnitude (Table A.1). Cygnus OB2 is not coeval. Its three subgroups have counts-based ages of 4.00, 4.09 and 2.52 Myr — **two older subgroups and one younger** — with a spread of 1.57 Myr between the central values. This differs from the structure the same pipeline returned before the extinction prior was corrected, which read as one older subgroup and two younger; the change followed a pre-registered fix whose four predictions were confirmed in advance.

The spread between three central values is not the age uncertainty. The envelope across both retained age indicators is 2.25–5.67 Myr, and that is what everything downstream inherits (Fig. 5). Within it, B’s posterior rails against its prior grid and is a one-sided bound, and C’s age is family-dependent in a way that moves it across the boundary discussed in Sect. 4.5.

4.3. Response-aware mass functions and the fit-quality map

On the baseline branch the mass function is well fitted in all three subgroups and the residual-fit criterion passes in each (Fig. 6, top). Across the full grid, 40 of 54 subgroup fit cells pass. That number alone invites a reading it does not support, so we give the failure map.

The 14 failures are distributed by subgroup as 4/7/3 for A/B/C, by isochrone family as 4 PARSEC against 10 MIST, by extinction law as 4/5/5 at $R_V = 3.0/3.1/3.5$, and by slope as 7/2/5 at $\alpha = 2.0/2.3/2.6$. They do *not* concentrate in the $R_V = 3.5$ and $\alpha = 2.6$ corners: the dominant axes are isochrone family and, on the headline arm, the shallow slope.

Three statements must therefore be kept apart. The baseline branch is accepted in all three subgroups. 40 of 54 cells pass the criterion. And of the 36 headline ledger branches, only 15 rest entirely on cells that passed — because within the headline set 27 of 36 cells pass, and only 5 of 12 family– R_V – α combinations pass in all three subgroups simultaneously. **The full grid did not pass the residual-fit criterion, and the headline set is not a validated subset of it.** The consequence for the cocoon application is quantified in Sect. 5.4.

4.4. The living massive-star census and subgroup-dependent closure

The strongest objection to any deficit method is that the IMF slope is constrained where the counts are complete and used where they are not. Here that objection has an answer, because the closure test never entered the fit.

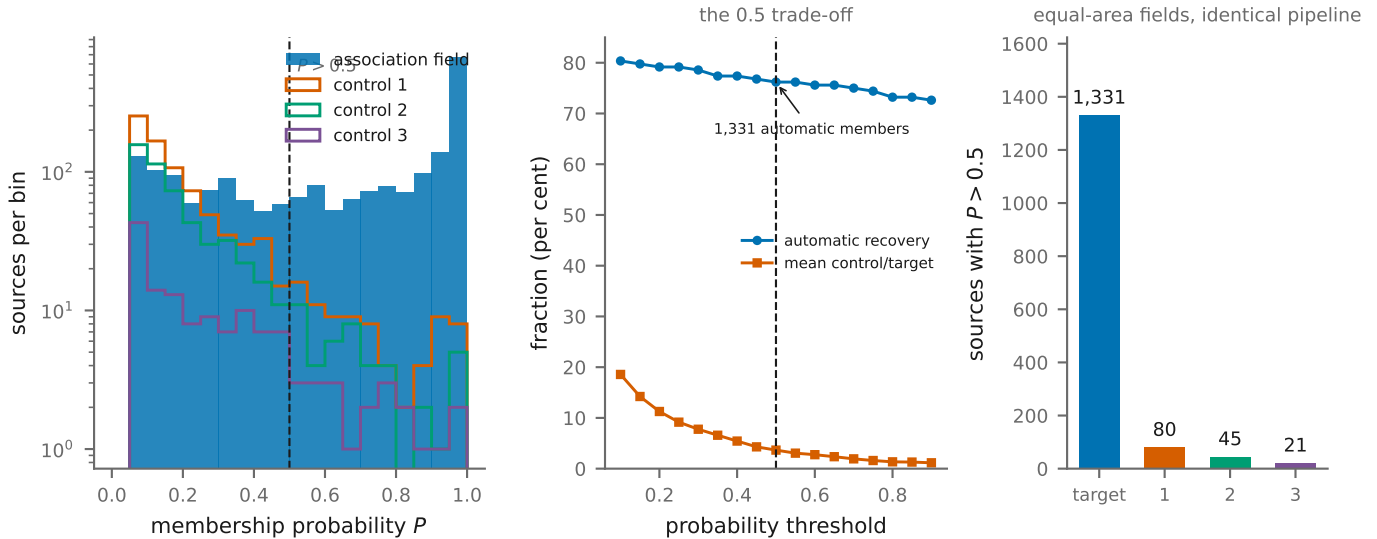


Fig. 4. Membership calibration against three control fields at matched Galactic latitude, processed through the identical selection, clustering and membership model. *Left:* membership-probability distributions on identical axes. *Centre:* external-list recovery and the mean control-to-target yield as the catalogue threshold changes; the dashed line is the adopted maximum-posterior boundary. *Right:* the numbers of clean automatic sources above $P > 0.5$ in the association and control fields. The control fields calibrated the mixture prior, so their yield measures internal consistency rather than a fully independent contamination rate.

The living census totals 380.6 stars above $8 M_{\odot}$: 295.4 in the member channel, 27 spectroscopic anchors inside the footprint but outside the member sample, and 58.2 runaways. The runaway search recovers 119 raw candidates, 54.9 after the measured false-positive correction, and it recovers the canonical ejected star BD+43°3654 at unit probability with 38.8 km s^{-1} and a 1.36 Myr look-back time against literature values of $\sim 40 \text{ km s}^{-1}$ and 1.6 Myr — from an estimator that knows nothing about the published answer.

Association-wide closure is near Salpeter, but the subgroups differ. Aggregated over the grid at $\alpha = 2.3$, observed and predicted counts above $8 M_{\odot}$ agree at the few-to-fifteen per cent level, and *which* number one quotes depends on the aggregation: the median of the 18 subgroup ratios is 1.067, while the star-weighted ratio of summed observed to summed predicted counts is 1.154. We report both because they are different quantities and because the difference is itself informative — the subgroups carry very different numbers of observed massive stars.

The residual is emphatically not uniform (Fig. 7). At $\alpha = 2.3$ the grid-median closure ratios are 0.95 for A, 1.07 for B and 1.39 for C — on the baseline branch alone, 0.865, 1.068 and 1.360. The slopes at which each subgroup’s census would close exactly are 2.34, 2.25 and 2.06, against 2.20 for the association aggregate and 2.25 as the median over the 18 individual subgroup cells, all 18 of which close inside the carried slope range rather than requiring an extrapolation. A and B want something close to Salpeter; C wants something distinctly shallower.

We therefore do *not* claim that a single power law is validated uniformly from 2 to $120 M_{\odot}$. What is validated is that the association-wide extrapolation closes near Salpeter, and what is exposed is that imposing one global slope forces the same α on three subgroups that do not want it. C’s excess is now the fourth independent appearance of a “Cyg OB2-C is different” signal, alongside its shallower preferred slope, its family-dependent age and its behaviour in the multiplicity test; the leading candidate remains distance contamination, which would inflate masses and

therefore closure ratios in exactly C’s direction. We report it as an open science question rather than absorbing it.

4.5. A branch-resolved supernova history

Under the permissive all-explode prescription, the baseline branch gives

$$N_{\text{SN}} = 8.43 \quad (\text{median } 8, 68\% [5, 11]), \quad (1)$$

comprising 4.17 from Cyg OB2-A, 4.26 from Cyg OB2-B and exactly 0.00 from Cyg OB2-C, whose turnoff on this branch has not yet fallen below the IMF’s upper limit. Within the same prescription, the probability that at least one supernova has occurred is 0.9997; the probability that the most recent falls within the last 100 kyr is 0.552, and the median time since it is 86 kyr. The first explosion was 1.30 Myr ago and the rate has been roughly flat at 8.0 Myr^{-1} since (Fig. 8).

The number is not 8.43. Across the 36 headline all-explode branches the total spans 5.63–28.7, a factor of 5.1; including $\alpha = 2.6$ it reaches down to 1.93, a factor of 14.9. The recent-event probability likewise spans 0.411–0.889. The branch spread is the uncertainty, and it exceeds the Poisson interval of any single branch. We quote the baseline value with its carried range beside it and never an average across branches, which for a set that carries no probability weights would not be a posterior summary.

4.6. What is robust about the explosion history

The whole budget lies in the very-massive regime. Every star that has died in Cygnus OB2 is very massive. Across the grid the lowest progenitor that explodes on any branch is $33.9 M_{\odot}$, and the lowest turnoff any star is compared against on a co-eval branch is $45.4 M_{\odot}$. Consequently, **for any direct-collapse threshold at or below $30 M_{\odot}$ the ledger returns exactly zero**

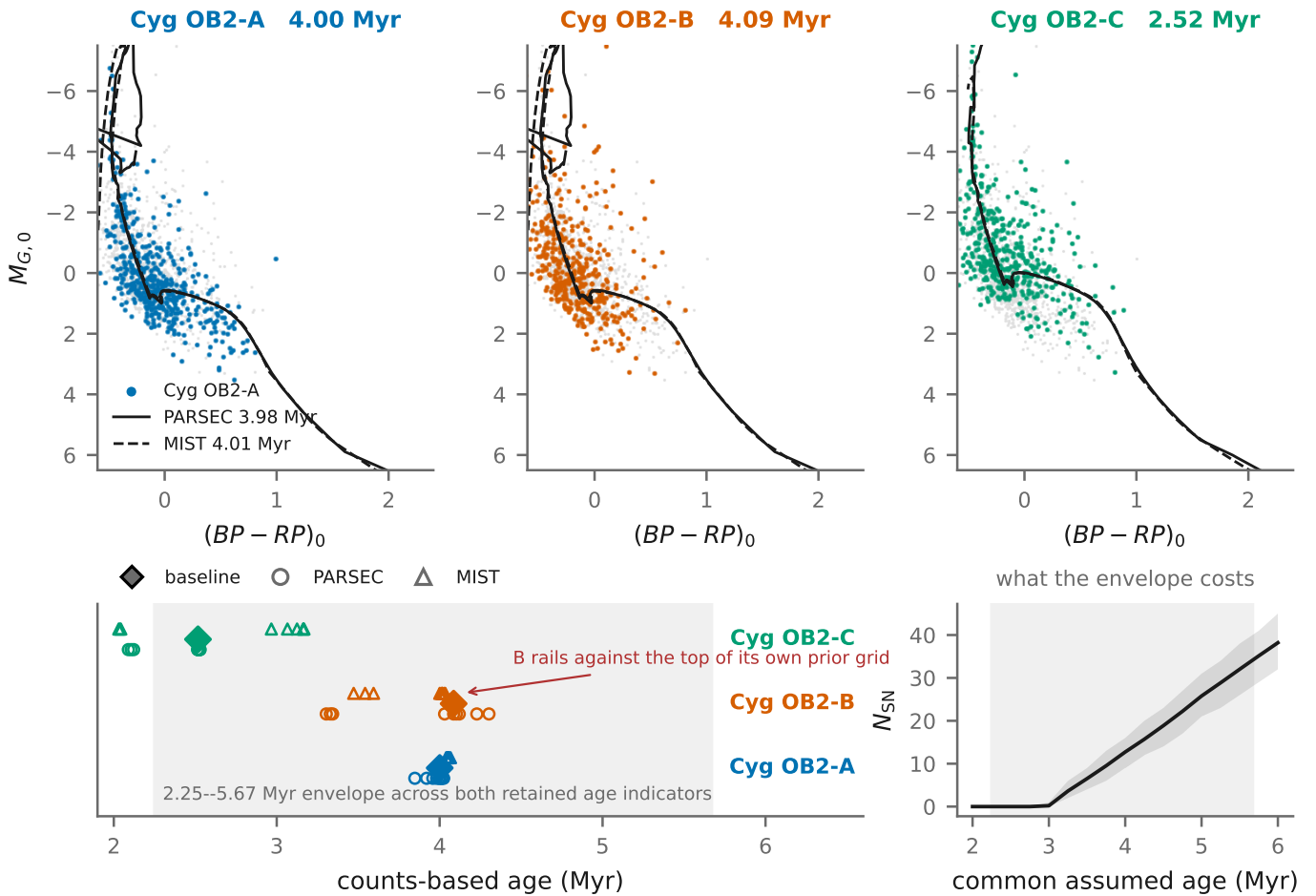


Fig. 5. The age evidence. *Top*: extinction-corrected colour–magnitude diagrams for the three subgroups on the baseline branch, with the other subgroups’ members in grey, spectroscopic anchors as open squares, and both isochrone families drawn at the adopted counts-based age of that subgroup. The observational scatter within a panel and the PARSEC-versus-MIST separation between the curves are different quantities and are drawn differently. *Bottom left*: the counts-based age of every fit cell, by subgroup and family, with the baseline branch marked and the full 2.25–5.67 Myr systematic envelope shaded; Cyg OB2-B’s cells pile up against the top of its own prior grid. *Bottom right*: what that envelope costs in supernovae.

510 **supernovae, on every branch and in every iteration.** The interleaved pattern of explodable and non-explodable ZAMS masses that Sukhbold et al. (2016) and Ertl et al. (2016) find between roughly 15 and 25 $M_{\odot}$ is irrelevant here, because nothing in that range has had time to die.

This is not a marginal systematic but the shape of the result: the stellar census measures the deaths available to produce supernovae, while the explosion ledger conditions on whether stars well above 30 $M_{\odot}$ explode at all (Fig. 9). The same separation will matter in other young associations whose turnoff remains in this uncertain very-massive regime.

530 What survives the branch structure. Two statements survive every all-explode branch: the model-branch spread exceeds the stochastic interval of any single branch, and every inferred death sits above the canonical core-collapse regime. A non-zero explosion count does *not* survive the explodability branches: the hard cutoff can reduce it to zero. The independent pulsar supplies the complementary fact that, under the association hypothesis, nature did not choose an explosion-free history. Together these results require at least one of three physical resolutions: a successful explosion in the very-massive regime, an older pro-

genitor population, or binary stripping of a lower-mass progenitor. The count itself, its recency, and — as Sect. 5.4 shows — its relevance to the cocoon remain conditional.

4.7. Independent evidence for past explosions

Every comparison in this subsection was pre-registered, with its statistic and pass condition fixed, before the ledger existed. All five predictions passed.

PSR J2032+4127 excludes an explosion-free history. The pulsar’s Be-star companion MT91 213 is *our own census star* (Gaia DR3 2067835682818358400), a B0V of 17 $M_{\odot}$ lying 0:115 from the Cyg OB2-A centroid and already counted in the living census. A neutron star requires a *successful* explosion. If the association is accepted, the pulsar therefore **independently excludes an entirely explosion-free history**: the ledger gives $P(\geq 1 \text{ SN}) = 0.9997$ if massive stars explode, against exactly 0 under a hard direct-collapse cutoff. Its characteristic age, widened for the line-of-sight acceleration its decades-long eccentric orbit imposes, gives 151–401 kyr, against a ledger probability of 0.960 that the last supernova falls inside that window;

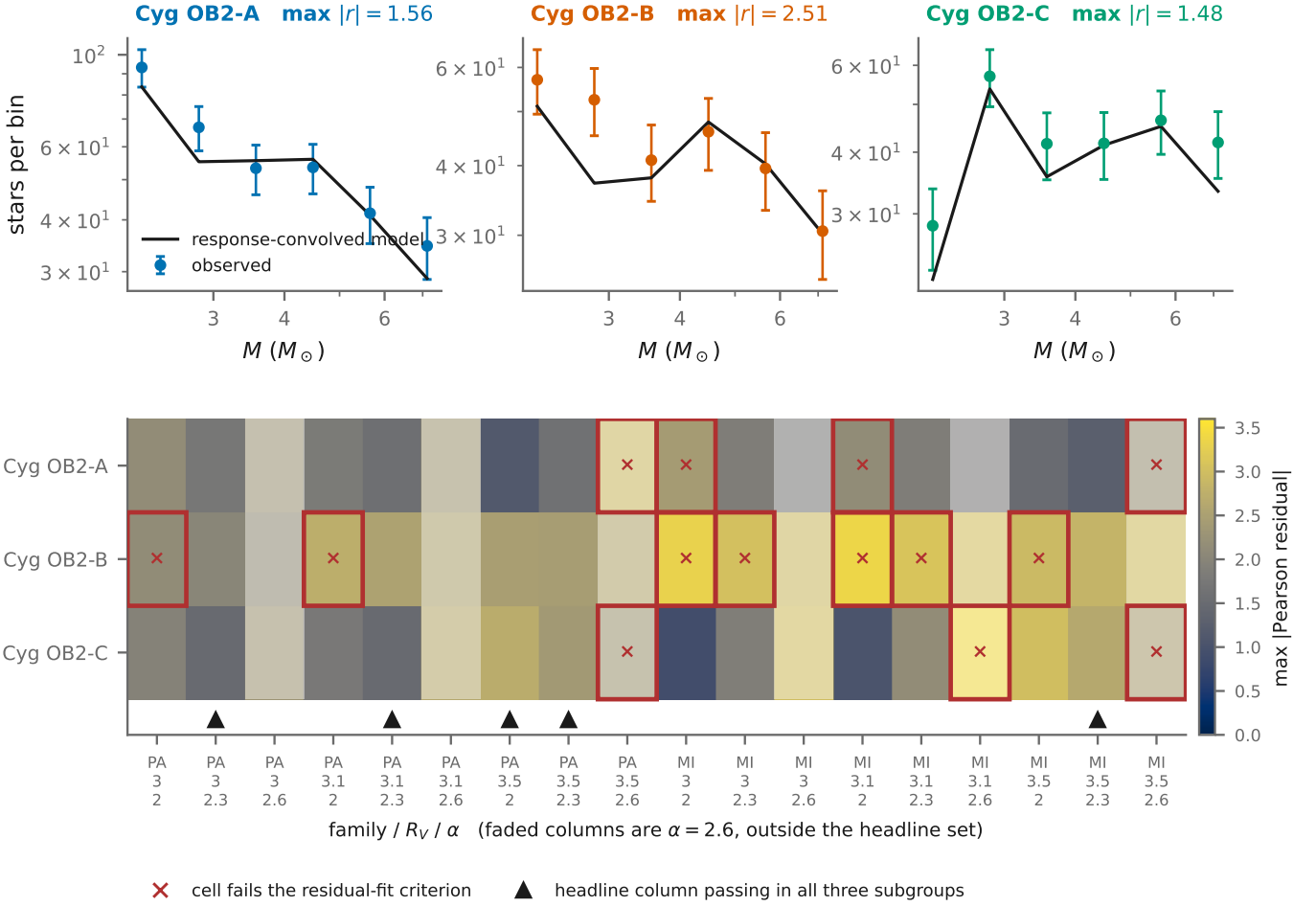


Fig. 6. Mass functions and the fit-quality landscape. *Top:* the calibration-window mass function per subgroup on the baseline branch against the response-convolved model — a Poisson likelihood against a forward model, not a fit to a completeness-divided histogram — with the worst Pearson residual annotated. *Bottom:* all 54 subgroup fit cells, coloured by worst absolute Pearson residual. Cells that fail the residual-fit criterion are outlined and crossed. Columns at $\alpha = 2.6$ are faded because they lie outside the headline set. A triangle marks the 5 headline columns that pass in all three subgroups.

its peculiar transverse velocity of 27.6 km s^{-1} is as low as a still-bound binary requires.

What the pulsar does *not* establish is that one of the $> 33.9 M_{\odot}$ single-star progenitors this ledger infers was the star that exploded. Three readings remain degenerate and we do not pretend otherwise: high-mass explodability, a population older than we fit, or binary stripping of a lower-mass progenitor. *All three give a non-zero budget*, which is what the argument needs and all that it needs.

γ Cygni — allowed, unsettled. The ledger gives $P = 7.7\%$ for a supernova within the remnant’s $\sim 7 \text{ kyr}$ age, so a physical association is allowed but not required. Our association distance of 1.62 kpc overlaps the remnant’s $1.7\text{--}2.6 \text{ kpc}$ range only at its low extreme. We leave this unsettled by design.

Absent remnants — weak evidence. Even at a generous 100 kyr visibility lifetime and with no cavity, the ledger predicts only 0.80 visible remnants. The absence of a catalogued remnant is therefore consistent with the cavity argument but does not test it.

^{26}Al — a consistency band. ^{26}Al traces both supernova ejecta and Wolf–Rayet winds, so it constrains the combination and cannot be inverted for a supernova count. It is reported as a band and never used as a fit. Run forward rather than backward — and, unlike everything else in this subsection, post-hoc rather than frozen at the outset (Sect. 5.8) — the ledger’s supernovae alone supply $10\text{--}42\%$ of the measured complex-wide 1809 keV flux (Martin et al. 2009). The supernova component is a subdominant but not negligible contributor to a measurement that also contains wind ^{26}Al and neighbouring populations; it is not the whole of it.

5. Discussion

5.1. A measured history, not a single supernova number

The result of this paper is a distribution over histories, not a number. Three kinds of uncertainty are kept separate throughout and should be read separately. The *statistical* uncertainty on the normalization is small. The *stochastic* uncertainty from sampling a discrete population at these numbers is the 68% interval $[5, 11]$ quoted with the baseline. The *model-branch* spread, $5.63\text{--}28.7$, is larger than either, and it is not a probability distribution: the

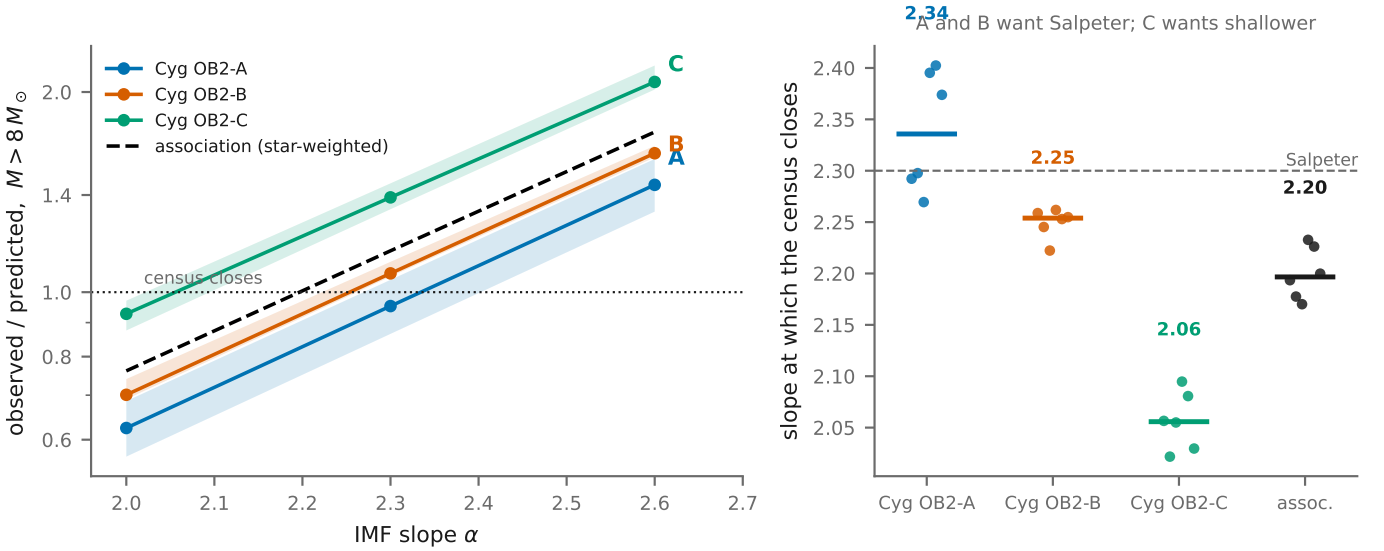


Fig. 7. The out-of-sample closure test, by subgroup and slope. *Left:* observed over predicted living stars above $8 M_{\odot}$ against α ; the line is the grid median over the six family- R_V cells and the band is their full range. The dashed curve is the star-weighted association aggregate. Unity is where the census closes. *Right:* the slope at which each cell's census closes exactly, one point per family- R_V cell, with the subgroup median marked. Cyg OB2-A and -B sit near Salpeter; Cyg OB2-C sits well below it.

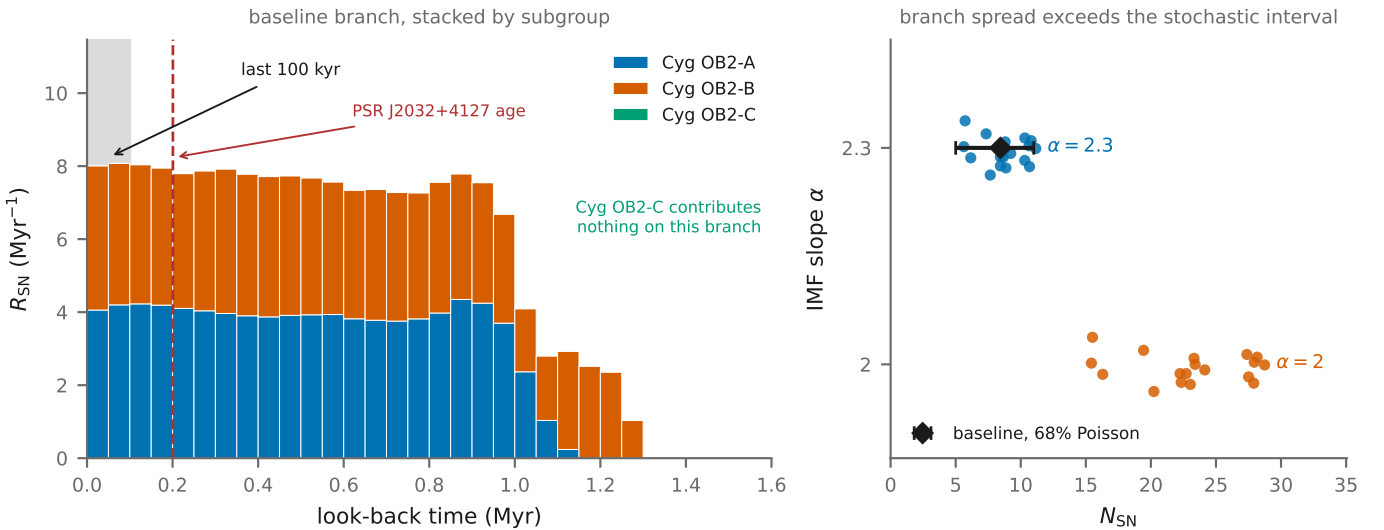


Fig. 8. The branch-resolved explosion history under the all-explode prescription. *Left:* $R_{\text{SN}}(t)$ on the baseline branch, stacked by subgroup, with the 100 kyr window shaded and PSR J2032+4127's characteristic age marked. *Right:* N_{SN} for every headline branch, separated by IMF slope, against the baseline value with its 68% Poisson interval. The discrete branch spread is the larger uncertainty and the two are not combined.

branches carry no weights, so their range is a statement about what we do not know rather than a posterior.

Quoting 8.43 alone would be defensible only if the branch axes were known to be unimportant. They are not: α alone moves the answer by a factor of several, and the age envelope moves it by more.

5.2. The very-massive-star explodability problem

The threshold scan of Fig. 9 is the primary explodability result, and its message is that this ledger is a probe of a regime where theory is least settled. The hard-cutoff branch we carry is exactly that — a hard cutoff — and we have renamed it accordingly throughout; it is not an implementation of the interleaved

explodability structure of Sukhbold et al. (2016) and Ertl et al. (2016), which in any case lies below the turnoff at these ages and cannot act.

The pulsar bounds this from one side: at least one star did explode. It does not tell us which, and the physical conclusion remains conditional on the fate of stars above roughly $30\text{--}50 M_{\odot}$.

5.3. Does one IMF describe all three subgroups?

Section 4.4 showed that they do not want the same slope. This matters twice over. First, the global- α grid imposes a common slope, so the branch range it produces understates the freedom the data actually leave. Second, Cyg OB2-C's excess, its shallower preferred slope, its family-dependent age and its possi-

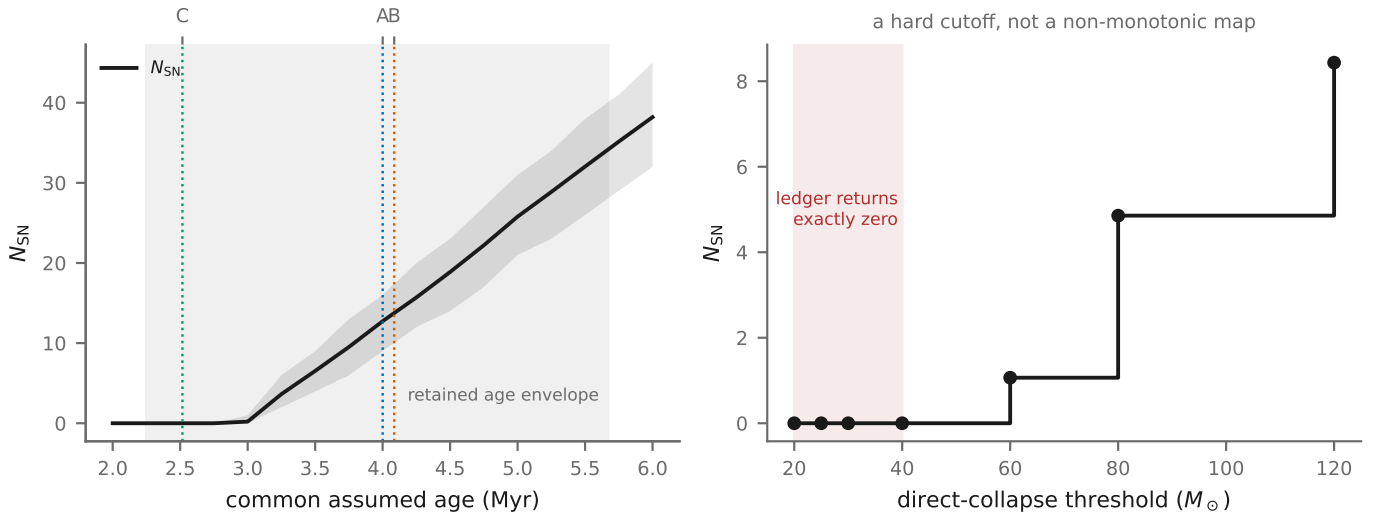


Fig. 9. The two sensitivities that dominate. *Left:* N_{SN} against a common age applied to all three subgroups, with the adopted subgroup ages marked on the upper axis and the retained 2.25–5.67 Myr envelope shaded. *Right:* N_{SN} against the hard direct-collapse threshold. Below 30 $M_{\odot}$ the ledger is identically zero. This is a threshold scan, not a non-monotonic explodability map.

ble distance contamination are one connected finding rather than four separate anomalies: all four are what a population contaminated by a slightly more distant group would look like.

We tested the first point directly by re-running the ledger with each subgroup assigned the carried slope its own closure prefers — A and B at $\alpha = 2.3$, C at $\alpha = 2.0$ on every family and R_V cell. The mixed-slope ledger spans 5.63–15.2 supernovae, which lies *inside* the carried headline range, so relaxing the single-slope assumption does not open a range beyond the one already reported. But its effect is strongly branch-dependent: on the 9 branches where C’s turnoff sits at the IMF ceiling the change is at most 1%, whereas on the 9 where C is dynamically alive it reaches 44%. The subgroup heterogeneity is therefore not presentational — where it acts, it is worth as much as an R_V step. The engine reproduces the stored single-slope ledger to 0.2%.

5.4. Application to the Cygnus cocoon

The ledger was measured without reference to any high-energy interpretation of the region; it can now be confronted with the interpretation that demands the most of it. Evaluating the requirement of Härer et al. (2025) — an event of the right age, progenitor type and location — over the 36 headline branches gives a score

$$S = C_1 C_3 C_4 = 0.323\text{--}0.735 \quad (\text{median } 0.533). \quad (2)$$

What S is, and what it is not. S is a *conditional scenario-availability score*, not a calibrated probability, and we do not present it as one. Only C_1 , the age term (0.379–0.861), is a sampled Monte-Carlo probability. C_3 is a deterministic mass-to-type mapping. C_4 is an upper bound held fixed. Independence of the three is asserted rather than demonstrated, and the explosion-energy condition is deliberately not multiplied in, because this analysis measures progenitor masses and not explosion energies. Table A.3 states the status of each term and its dominant limitation.

C_3 is a model mapping. $C_3 = 1.000$ exactly, because every simulated progenitor exceeds the adopted 30 $M_{\odot}$ stripping threshold. That is a fact about the ledger’s mass floor, not evidence that every explosion would have been observed as a stripped-envelope event. Whether a star of a given ZAMS mass arrives at collapse without its hydrogen envelope depends on its wind mass-loss history, which is metallicity-dependent (Vink et al. 2001) and whose absolute scale has been revised downwards substantially by more recent theory (Björklund et al. 2021) and by observational reassessments of clumping (Smith 2014), and on binary interaction, which affects the majority of massive stars (Sana et al. 2012; de Mink et al. 2014). We therefore report C_3 as a bracket rather than a measurement: moving the threshold to 40 $M_{\odot}$ gives $C_3 = 0.98\text{--}1.00$, and an explicitly pessimistic 60 $M_{\odot}$ assumption gives 0.68–0.96. Under that pessimistic mapping S spans 0.304–0.691, and 18 of 18 shallow-slope branches still clear the 0.5 boundary — so the qualitative split survives, but $C_3 = 1.000$ does not survive as a statement about nature.

C_4 is a bound, so we show the answer against it. C_4 is fixed at 0.854 from the runaway census. That census is two-dimensional and footprint-limited, so it is a lower bound on escape and hence an *upper* bound on the in-situ fraction; and it is measured on living stars below the turnoff, whereas the quantity required is the escape fraction of already-dead, more massive progenitors, which need not be the same. Figure 10 therefore shows S as a function of C_4 rather than at a point.

Two thresholds must be stated separately, and the distinction is not cosmetic. C_4 below 0.720 moves at *least one* shallow-slope branch below 0.5; C_4 below 0.581 moves *every* one of them below it. In the other direction, C_4 would have to exceed 0.901 before even the highest $\alpha = 2.3$ branch rose above 0.5 — so the split between the two slopes is not an artefact of the adopted C_4 , though it is not immune to it either.

The score hinges on one axis. The spread in S is 0.264 across α , against 0.065 across R_V , 0.045 across isochrone family and 0.043 across star-formation duration. At the adopted C_4 the split

is exact: every one of the 18 branches at $\alpha = 2.0$ clears 0.5 (0.593–0.735) and none of the 18 at $\alpha = 2.3$ does (0.323–0.474), with the boundary falling between the two slopes. The sensitivity-only $\alpha = 2.6$ branches give $S = 0.142$ –0.250.

The split, restricted to validated fits. The headline result — every $\alpha = 2.0$ branch above 0.5, no $\alpha = 2.3$ branch above it — uses branches that include failed mass-function cells (Sect. 4.3). Restricting to the 15 branches built entirely on cells that passed, the split survives: 3 of 3 remaining $\alpha = 2.0$ branches stay above 0.5 (0.685–0.721) and 0 of 12 remaining $\alpha = 2.3$ branches rise above it. The qualitative conclusion is robust; the branch representation is badly imbalanced, and the “all versus none” phrasing should be read with that in mind.

What we cannot say. The analysis estimates the *availability* of a suitable event from Cygnus OB2. It cannot quantify whether Cygnus OB2, rather than a neighbouring Cygnus population, was the source of an event somewhere in the larger cavity. We attempted a coarse literature-based budget for the neighbouring associations from the tabulation of Martin et al. (2010) and Knödseder et al. (2002), with a pre-registered validation test requiring that the same coarse estimator reproduce our own measured ledger to within an order of magnitude when applied to their Cyg OB2 row. **That test failed**, and we report it as failed: at the 2.5 Myr age those authors adopt for Cygnus OB2 the turnoff lies above the IMF ceiling and the coarse estimator returns identically zero against our measured 8.43. Run instead at the age we measure it overshoots by a factor 2.1–3.9. The estimator is dominated by the assumed age, which for the neighbours is exactly the quantity no homogeneous census yet constrains. We therefore decline to publish a neighbour budget rather than publish a number rejected by its own validation test, and the cocoon score is to be read as Cygnus OB2’s availability alone. Appendix E records the attempt.

The framing rule. The rule — fixed at planning time and operationalized before the number was known — returns INCONCLUSIVE, and hence this article rather than a Letter. Under the model’s own permissive “few hundred kyr” allowance every branch clears 0.5 ($S = 0.727$ –0.854). The scenario is therefore *available* against its authors’ permissive statement and *unresolved* against their preferred ~ 50 kyr. Both windows were pre-registered before either was computed.

Near one half is not uninformative. The score changes strongly with the physically specified time window: it is 0.055 for 7 kyr, 0.552 for 100 kyr and 0.854 for the permissive window. A last supernova distributed uniformly over the association’s 1.30 Myr active period would fall inside 100 kyr with probability 0.077; we measure 0.552, a factor of 7 higher, because the rate is measured rather than assumed.

5.5. Binary mass transfer: a conservative sensitivity bracket

Our count is a single-star turnoff calculation, and most massive stars have companions — Sana et al. (2012) find a close-binary fraction near 0.7 for 15–60 $M_{\odot}$, and about two-thirds of Cygnus OB2’s own O and early-B stars are resolved or spectroscopic multiples. Two independent lines indicate what that omission is worth.

Empirically, the BPASS (Stanway & Eldridge 2018) calculation of Härer et al. (2025) — *with* binaries, for a population of the same mass and IMF slope — gives a core-collapse rate at 4 Myr that is 0.72–0.73 times ours once the difference in IMF integration ranges is corrected: the binary-inclusive rate sits *below* our single-star one. Theoretically, Zapartas et al. (2017) find that binarity raises the integrated number of core-collapse supernovae by $14^{+15}_{-14}\%$, but attribute the increase to *late* events 50–200 Myr after birth from 4–8 $M_{\odot}$ binaries, and find the early delay-time distributions “remarkably similar”, diverging only around 20 Myr. Cygnus OB2 is 4 Myr old, so the entire ledger sits inside the regime where the two distributions agree.

Adopting a deliberately conservative $\pm 30\%$ bracket — the top of the Zapartas et al. (2017) enhancement applied in full despite its channel being inapplicable — and running it through the ledger engine moves the baseline from 5.90 to 10.96 supernovae, a span of 5.1 against a headline branch span of 23.1. **This is a scenario bracket, not a formal statistical bound on binary evolution at 4 Myr**, and we do not claim the unmodelled binary systematic has been bounded in a strict probabilistic sense. What it does establish, and what matters here, is that the tested bracket is about 5 times smaller than the model-branch spread we already report.

Nor does the living binary fraction translate directly. That $30^{+10}_{-15}\%$ of massive main-sequence stars are binary products (de Mink et al. 2014) is a statement about survivors: accretors and mergers are rejuvenated, so a star pushed above the present turnoff by mass transfer has had its clock partly reset and has typically *not* yet exploded.

5.6. Systematics budget

The absolute extinction scale. The anchor-versus-photometry offset of Appendix C is a systematic on the mass *scale*: it moves all three subgroups together and does not produce the A/B/C ordering. Everything that scales with mass — the normalizations $k = 1, 734, 1,647$ and $1,894$, the turnoff masses 58, 56 and 279 $M_{\odot}$, and N_{SN} — inherits it.

Cyg OB2-B’s age is a one-sided bound. 53% of B’s counts-based age posterior sits on the topmost of its nine prior nodes. A posterior that piles up against the edge of its prior grid reports a bound, not a measurement, and the direction is known: were B’s true age the top node (4.26 Myr) its supernova count would be higher still. B’s contribution to N_{SN} is a lower bound at the precision quoted.

The in-situ fraction is an upper bound. Of the living census, 322.4 stars above 8 $M_{\odot}$ were retained in the association and at most 14.6% of the ledger’s supernovae were not in situ. Because the runaway count is a lower bound (Sect. 3.8), the true escaped fraction is higher and C_4 correspondingly lower — a direction that works against the cocoon scenario, quantified in Fig. 10.

The age dependence, disclosed rather than absorbed. Below 3.00 Myr the turnoff has not crossed the IMF ceiling and nothing has died; above it the count rises steeply, reaching 38.2 supernovae at 6 Myr (Fig. 9). Every number in this paper inherits the ages of Sect. 3.5, and that is the sensitivity.

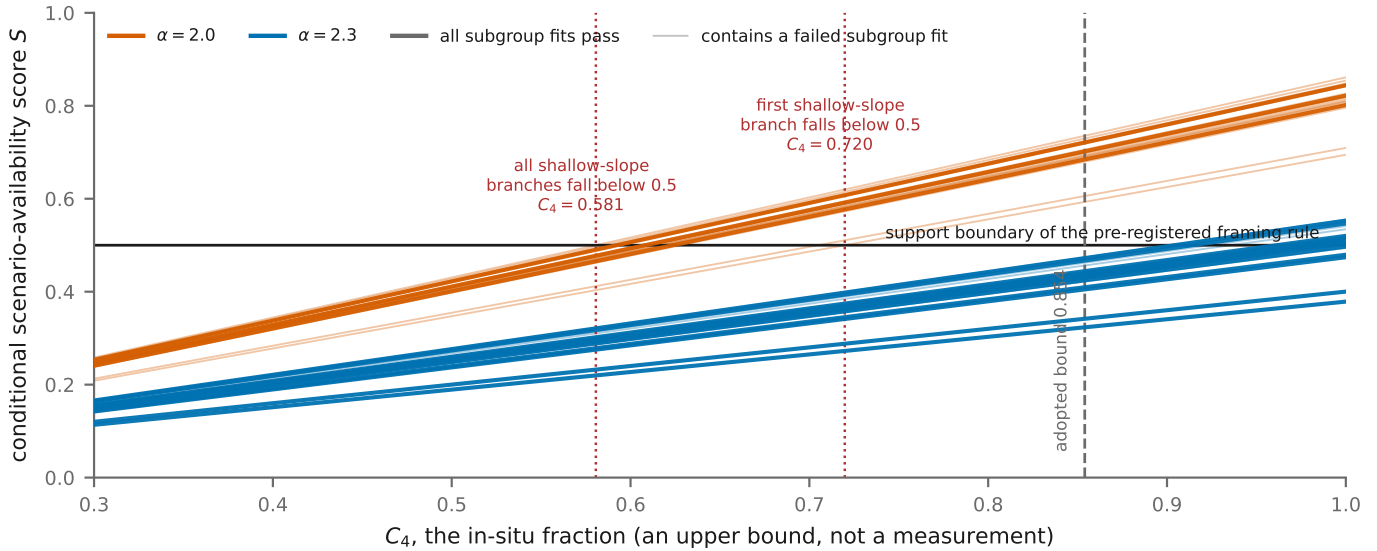


Fig. 10. The conditional scenario-availability score against C_4 , for every headline branch. Thick lines are branches whose mass-function fits pass the residual-fit criterion in all three subgroups; thin faded lines contain at least one failed cell. The dashed vertical line is the adopted C_4 bound; the dotted lines mark where the first and then every shallow-slope branch falls below the 0.5 boundary of the pre-registered framing rule. The ordinate is a scenario score, not a probability.

Where the deaths sit. Across the whole grid the lowest turnoff any star is compared against is $33.9 M_{\odot}$ and the lowest exploding progenitor is $33.9 M_{\odot}$, reached only on the widest star-formation-window branches, where up to 19% of the supernovae come from progenitors below $52 M_{\odot}$. On the baseline and on every coeval branch the floor is higher.

Association mass, and which one. Three nested integrals of the same normalization can all be called “the association mass”, and they differ by a factor of 1.67. We quote the multiplicity-adjusted stellar mass over $0.08\text{--}120 M_{\odot}$, $2.92 \times 10^4 M_{\odot}$. The like-for-like partner of Wright et al.’s 2015 $1.65 \times 10^4 M_{\odot}$ — primary stars over the full range — is $2.42 \times 10^4 M_{\odot}$, a factor 1.47. Integrating only $0.5\text{--}120 M_{\odot}$ instead gives $1.75 \times 10^4 M_{\odot}$, which coincides with the literature value but compares a different quantity. The comparison is an agreement at the factor level, which is the resolution IMF-extrapolated cluster masses admit.

5.7. Pre-registration as part of the result

Every marker comparison in Sect. 4.7 was pre-registered, with its statistic and pass condition fixed, before the ledger it tests existed; the markers themselves were frozen at data acquisition. The question “did you tune the analysis to the markers?” therefore has a checkable answer rather than a reassurance. The same discipline produced results we would rather not have: predictions recorded as failed and not reinterpreted, two intermediate results withdrawn after defects were found by comparison against externally known values, and the neighbouring-population validation test of Sect. 5.4 (Appendix G). We report this not as virtue but because the score’s credibility rests on it: an analysis that can be tuned to a ~ 50 kyr window is not evidence about a ~ 50 kyr window.

5.8. Radioactive-isotope consistency and forecast

Unlike the cross-checks of Sect. 4.7, this comparison was not frozen before the analysis began: it is a post-hoc addition, chosen after the ledger existed and pre-registered before it was scored. We flag it because the weight the frozen markers carry depends on that distinction.

^{60}Fe is the supernova-specific tracer — Wolf-Rayet winds make none of it, so unlike ^{26}Al it counts explosions rather than a combination. Carrying published per-supernova yields as a branch fixed before any flux was computed, and accumulating each event’s ejecta at its own progenitor mass with decay from its own epoch, the headline branches predict $3.0\text{--}14.3 \times 10^{-3} M_{\odot}$ of ^{60}Fe surviving now, beside $0.92\text{--}3.7 \times 10^{-3} M_{\odot}$ of supernova ^{26}Al . The association is nowhere near steady state — only a fraction 0.16–0.31 of the saturated ^{60}Fe mass has accumulated — so the usual rate-times-lifetime estimate would overstate the signal several-fold.

A COSI measurement or upper limit would constrain the combination of high-mass IMF slope and very-massive-star yield and explodability, not α alone. That joint statement is the honest form of the forecast, and Fig. 11 shows why: the spread between published yield arms is a factor 30 in the predicted flux, against 4.8 across the entire headline branch set (Limongi & Chieffi 2006). Within the primary arm the prediction is nevertheless clean: at 1.62 kpc the median 1173 keV flux is $3.6 \times 10^{-6} \text{ ph cm}^{-2} \text{ s}^{-1}$ against COSI’s expected 3σ narrow-line sensitivity of 3.0×10^{-6} in a two-year survey (Tomsick & COSI Collaboration 2023), and 18 of 18 branches at $\alpha = 2.0$ clear it while 0 of 18 at $\alpha = 2.3$ do, a factor 2.68 between the medians.

Two things bound that claim and we give them equal weight. The current standard compilation predicts nothing at all: Limongi & Chieffi (2018)’s recommended scenario collapses every star above $25 M_{\odot}$ straight to a black hole, and 0 of the 315 million supernovae we sample fall below that threshold, so on that prescription this ledger’s ^{60}Fe yield is identically zero (Falla et al. 2025). A non-detection would then favour precisely

the explodability that PSR J2032+4127 disfavours, and that tension would itself be the result. And existing data are already near the boundary: our richest branch predicts a combined ^{60}Fe flux within 5% of the INTEGRAL/SPI upper limit for the Cygnus region, $1.6 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$ (Martin et al. 2009).

The forecast is ^{60}Fe -rich by Galactic standards — the predicted $^{60}\text{Fe}/^{26}\text{Al}$ flux ratio is 0.76–1.00, against 0.184 ± 0.042 measured for the Galaxy (Wang et al. 2020) — but the two are not the same quantity, and the difference is expected rather than anomalous: our ^{26}Al denominator counts supernova ejecta only, where the Galactic ratio’s includes wind ^{26}Al , and every progenitor here lies above $30 M_{\odot}$, where the ^{60}Fe yield peaks.

5.9. What Gaia DR4 will and will not resolve

DR4 improves three things that matter here: radial velocities for bright OB stars make the traceback three-dimensional, so runaways stop being a lower bound and C_4 tightens; better astrometry and photometry sharpen the subgroup ages, including Cyg OB2-C’s family-dependent ambiguity and the 0-to-several supernova range that follows from it; and deeper membership assigns subgroups to the currently unlabelled massive members and orphan anchors. *Gaia* XP spectra address Cyg OB2-B’s anchor famine specifically. A parallax-blind membership rerun also becomes far more informative at DR4 precision.

DR4 will not settle the dominant term. The score hinges on the high-mass IMF slope, with a spread of 0.264 against 0.045 for isochrone family, 0.065 for R_V and 0.043 for star-formation duration. Distinguishing $\alpha = 2.0$ from 2.3 needs a larger or deeper massive-star census, or a second association analysed identically — not better astrometry on this one.

5.10. Limitations and reproducibility

Three formal validation tests failed and are recorded as failed rather than reinterpreted. The ledger’s pre-registered *relative* convergence criterion failed; the stored execution record carries that verdict unchanged, and Appendix D reports the absolute drift of material cells, which demonstrates numerical stability without rewriting the criterion as a pass. One pre-registered ledger prediction about the mass floor failed and caught a real over-generalization. And the neighbouring-population validation test of Sect. 5.4 failed, with the consequence stated there.

Three questions are named and left for later work rather than held against this paper: a parallax-blind membership test, which is the only way to break the circularity in the two-distance non-confirmation; a distance-contamination test for Cyg OB2-C; and a quantitative neighbouring-association budget, which needs a homogeneous census of those associations rather than a coarser method applied to 2002-era literature counts.

Appendix F gives the reproducibility map: the versioned input list, the script-to-output map, the validation records, the superseded artifacts and why they are forbidden, the environment, and the command sequence that regenerates every number, table and figure in this paper.

6. Conclusions

1. Three stable kinematic subgroups are found within the selected Cygnus OB2 member sample, at 1.62 kpc. There is no evidence for two distinct distances *among the members we select*; because parallax was a selection feature, this does not exclude a foreground population.

2. Differential extinction is spatially coherent and cannot be represented by a single regional value, and the subgroup age structure — 4.00, 4.09 and 2.52 Myr, two older and one younger — materially changes the high-mass history. Cyg OB2-B’s age rails against its prior grid, so its supernova contribution is a one-sided lower bound.
3. The baseline supernova ledger is $N_{\text{SN}} = 8.43$ (median 8, 68% [5, 11]) under the all-explode prescription, but the honest conditional result is the carried range 5.63–28.7 across 36 all-explode branches. The model-branch spread exceeds the stochastic interval of any one branch.
4. All inferred deaths occupy the uncertain very-high-mass explodability regime: for any direct-collapse threshold at or below $30 M_{\odot}$ the ledger returns exactly zero, so the result is conditional on whether very massive stars explode, and the 15–25 $M_{\odot}$ island structure is irrelevant at this age.
5. PSR J2032+4127 independently excludes an entirely explosion-free history, if its association with Cygnus OB2 is accepted. Combined with the hard- cutoff result, this requires either a successful very-massive-star explosion, an older progenitor population, or a binary-modified lower-mass progenitor; it does not identify which channel occurred.
6. Association-wide closure is near Salpeter-like — 1.067 as a median of subgroup ratios, 1.154 star-weighted — but the subgroups are heterogeneous, with closing slopes 2.34, 2.25 and 2.06. A single universal slope from 2 to 120 $M_{\odot}$ cannot be claimed without that qualification.
7. The cocoon application is a conditional scenario-availability score ($S = 0.323\text{--}0.735$), not a calibrated probability. It is dominated by the IMF slope and rests on a deterministic type mapping and an in-situ upper bound. Whether Cygnus OB2 rather than a neighbour supplied an event elsewhere in the cavity is outside what these data constrain.
8. *Gaia* DR4 will improve ages, distance contamination and three-dimensional runaways; it will not by itself determine the high-mass IMF.
9. COSI can constrain the combined IMF, yield and explodability problem, with a clean split by slope on one specified yield arm.

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Data availability

The analysis code, the versioned intermediate tables, the pre-registration records and the validation records that reproduce every number in this paper are released with it; Appendix F gives the map from script to output and the command sequence. The bulk *Gaia* and 2MASS working tree is regenerated from the archival queries recorded there rather than redistributed.

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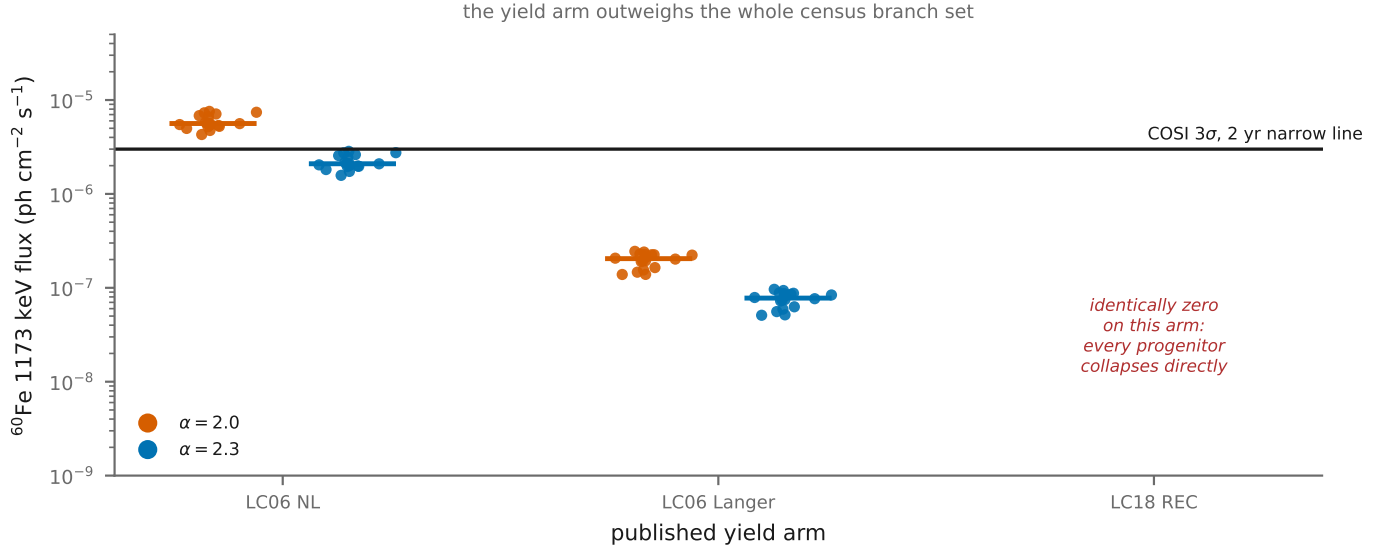


Fig. 11. The ^{60}Fe forecast against COSI’s expected sensitivity, for every headline branch, on each of the three published yield arms fixed before any flux was computed. The separation between arms is far larger than the spread across the whole census branch set, which is why a measurement constrains the IMF and the yield model jointly rather than the IMF alone. On the recommended arm of [Limongi & Chieffi \(2018\)](#) every progenitor collapses directly and the predicted flux is identically zero.

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Appendix A: Model branches and validation outcomes

1010 Table A.4 lists every all-explode branch with its mass-function fit status per subgroup, its closure ratio, its supernova count and interval, its recent-event probability and its conditional scenario score. Table A.2 summarizes every scientific analysis stage and the consequence carried downstream. The complete machine-readable table, with the per-subgroup closure ratios, the black-hole-threshold columns, the minimum turnoff and progenitor masses and the full Poisson quantiles, is published electronically.

1020 Two conventions in that table matter. A branch marked as failing the residual- fit criterion in one or more subgroups is *retained*, not deleted: it is an explicit sensitivity case, and dropping it because it fits badly under one assumption would be a way of selecting that assumption. And the closure ratio is observed over predicted, so values above unity mean the census contains more massive stars than the extrapolation predicts.

Table A.1. The three kinematic subgroups on the baseline branch (PARSEC, $R_V = 3.1$, $\alpha = 2.3$, $\delta = 0$). The age is the counts-based age the ledger consumes, not the colour–magnitude prior. Closure is the grid median over the six family– R_V cells at $\alpha = 2.3$, and the closing slope is the slope at which that subgroup’s census would close exactly. A $\times$ in the fit-status column marks a subgroup whose baseline mass-function fit failed the residual-fit criterion; none does.

subgroup	N	median A_V (mag)	age (Myr)	k	closure at $\alpha = 2.3$	closing α	N_{SN}	fit
Cyg OB2-A	476	5.50	4.00	1,734	0.95	2.34	4.17	✓
Cyg OB2-B	426	6.03	4.09	1,647	1.07	2.25	4.26	✓
Cyg OB2-C	429	6.44	2.52	1,894	1.39	2.06	0.00	✓

Table A.2. Validation status for each scientific analysis stage and the consequence carried downstream. PARTIAL means the stage was usable while a named criterion or limitation remained; the failure is reported in the text and was not reinterpreted.

analysis stage	verdict	consequence
Prior-art and scope review	PASS	no direct duplicate; one partial overlap cited
Data acquisition and marker freeze	PASS	catalogues recorded; external markers fixed before the ledger
Membership and kinematic structure	PASS	all eight acceptance criteria met after one rejected trial
Per-star extinction mapping	PASS	accepted after repairing the sparse-anchor treatment
Ages and stellar masses	PARTIAL	Cyg OB2-B’s age and supernova contribution remain one-sided
IMF normalization	PARTIAL	baseline passes in all three subgroups; 40 of 54 cells pass
Massive-star closure and runaways	PASS	both validation criteria met; three earlier defects withdrawn
Supernova ledger	PARTIAL	three of four validation criteria met; relative convergence failed
Independent supernova cross-checks	PASS	four criteria and five pre-registered predictions met
Pre-registered framing rule	PASS	rule returned INCONCLUSIVE mechanically
Radioactive-isotope forecast	PARTIAL	post-hoc; one test was vacuous and its corrected form failed
Fit-landscape and scenario sensitivities	PARTIAL	neighbouring-population calibration failed and scoped that claim out
Manuscript and reproducibility	PASS	generated numbers, source validation and clean PDF build pass

Table A.3. The three factors of the conditional scenario-availability score $S = C_1 C_3 C_4$, with what each one actually is. Only C_1 is a sampled probability. The product is therefore a scenario score, not a calibrated probability, and the paper does not present it as one.

term	what it is	numerical treatment	dominant limitation
C_1	sampled Monte-Carlo probability	$P(\geq 1 \text{ explosion in the required age window}), \text{ per branch}$	inherits the age and IMF branch structure; spans a factor of two across the headline set
C_3	deterministic model mapping	indicator that the progenitor exceeded $30 M_\odot$	returns exactly unity because the ledger’s mass floor lies above the threshold; falls to 0.68 under a $60 M_\odot$ assumption
C_4	upper bound	held fixed at 0.854	a two-dimensional, footprint-limited runaway census of <i>living</i> stars below the turnoff, applied to dead and more massive progenitors

Table A.4. The 54 all-explode branches, their mass-function fit-quality status and their results. Family PA = PARSEC, MI = MIST. δ is the star-formation duration in Myr. Columns A, B, C give the residual-fit verdict for each subgroup at that family, R_V and α ; *all* is their conjunction. Σ is the star-weighted association closure ratio. S is the conditional scenario-availability score of Sect. 5.4, *not* a calibrated probability. Branches with $\alpha = 2.6$ are sensitivity cases outside the headline set. The complete table, including per-subgroup closure ratios, black-hole-threshold columns and Poisson intervals, is available electronically.

fam.	R_V	α	δ	A	B	C	all	Σ	N_{SN}	68%	P_{100}	S
MI	3	2	0	×	×	✓	×	0.77	27.51	[22, 33]	0.87	0.724
MI	3	2	1	×	×	×	×	0.77	27.38	[22, 33]	0.87	0.721
MI	3	2	2	×	×	✓	×	0.77	28.16	[23, 34]	0.84	0.689
MI	3.1	2	0	×	×	✓	×	0.78	27.94	[23, 33]	0.86	0.706
MI	3.1	2	1	×	×	✓	×	0.78	27.90	[23, 33]	0.87	0.719
MI	3.1	2	2	×	×	✓	×	0.78	28.74	[23, 34]	0.85	0.696
MI	3.5	2	0	✓	×	✓	×	0.79	19.44	[15, 24]	0.73	0.593
MI	3.5	2	1	✓	×	✓	×	0.79	20.23	[16, 25]	0.75	0.606
MI	3.5	2	2	✓	×	✓	×	0.79	22.71	[18, 28]	0.83	0.680
PA	3	2	0	✓	×	✓	×	0.72	23.40	[18, 29]	0.84	0.691
PA	3	2	1	✓	×	✓	×	0.72	23.31	[18, 29]	0.84	0.692
PA	3	2	2	✓	×	✓	×	0.72	24.14	[18, 30]	0.88	0.729
PA	3.1	2	0	✓	×	✓	×	0.73	22.33	[17, 27]	0.85	0.699
PA	3.1	2	1	✓	×	✓	×	0.73	22.24	[17, 27]	0.85	0.700
PA	3.1	2	2	✓	×	✓	×	0.73	23.03	[18, 28]	0.89	0.735
PA	3.5	2	0	✓	✓	✓	✓	0.76	15.43	[11, 20]	0.87	0.721
PA	3.5	2	1	✓	✓	✓	✓	0.76	15.51	[11, 20]	0.85	0.702
PA	3.5	2	2	✓	✓	✓	✓	0.76	16.30	[12, 21]	0.84	0.685
MI	3	2.3	0	✓	×	✓	×	1.16	10.31	[7, 14]	0.59	0.474
MI	3	2.3	1	✓	×	✓	×	1.16	10.31	[7, 14]	0.58	0.462
MI	3	2.3	2	✓	×	✓	×	1.16	10.80	[7, 14]	0.53	0.422
MI	3.1	2.3	0	✓	×	✓	×	1.18	10.64	[7, 14]	0.56	0.443
MI	3.1	2.3	1	✓	×	✓	×	1.18	10.68	[7, 14]	0.57	0.456
MI	3.1	2.3	2	✓	×	✓	×	1.18	11.16	[8, 15]	0.55	0.433
MI	3.5	2.3	0	✓	✓	✓	✓	1.19	7.34	[5, 10]	0.41	0.323
MI	3.5	2.3	1	✓	✓	✓	✓	1.19	7.66	[5, 10]	0.44	0.342
MI	3.5	2.3	2	✓	✓	✓	✓	1.19	8.66	[6, 12]	0.52	0.410
PA	3	2.3	0	✓	✓	✓	✓	1.10	8.79	[6, 12]	0.54	0.433
PA	3	2.3	1	✓	✓	✓	✓	1.10	8.80	[6, 12]	0.54	0.430
PA	3	2.3	2	✓	✓	✓	✓	1.10	9.23	[6, 13]	0.59	0.467
★ PA	3.1	2.3	0	✓	✓	✓	✓	1.11	8.43	[5, 11]	0.55	0.440
PA	3.1	2.3	1	✓	✓	✓	✓	1.11	8.44	[5, 11]	0.55	0.437
PA	3.1	2.3	2	✓	✓	✓	✓	1.11	8.86	[6, 12]	0.59	0.472
PA	3.5	2.3	0	✓	✓	✓	✓	1.15	5.63	[3, 8]	0.56	0.444
PA	3.5	2.3	1	✓	✓	✓	✓	1.15	5.72	[3, 8]	0.53	0.424
PA	3.5	2.3	2	✓	✓	✓	✓	1.15	6.16	[4, 9]	0.51	0.405
MI	3	2.6	0	✓	✓	✓	✓	1.75	3.71	[2, 6]	0.32	0.250
MI	3	2.6	1	✓	✓	✓	✓	1.75	3.76	[2, 6]	0.30	0.232
MI	3	2.6	2	✓	✓	✓	✓	1.75	4.02	[2, 6]	0.26	0.205
MI	3.1	2.6	0	✓	✓	×	×	1.79	3.95	[2, 6]	0.28	0.219
MI	3.1	2.6	1	✓	✓	×	×	1.79	3.99	[2, 6]	0.29	0.226
MI	3.1	2.6	2	✓	✓	×	×	1.79	4.24	[2, 6]	0.28	0.215
MI	3.5	2.6	0	×	✓	×	×	1.79	2.70	[1, 4]	0.18	0.142
MI	3.5	2.6	1	×	✓	×	×	1.79	2.81	[1, 4]	0.21	0.158
MI	3.5	2.6	2	×	✓	×	×	1.79	3.20	[1, 5]	0.25	0.195
PA	3	2.6	0	✓	✓	✓	✓	1.67	3.13	[1, 5]	0.28	0.215
PA	3	2.6	1	✓	✓	✓	✓	1.67	3.15	[1, 5]	0.27	0.210
PA	3	2.6	2	✓	✓	✓	✓	1.67	3.37	[1, 5]	0.29	0.230
PA	3.1	2.6	0	✓	✓	✓	✓	1.68	3.06	[1, 5]	0.28	0.218
PA	3.1	2.6	1	✓	✓	✓	✓	1.68	3.09	[1, 5]	0.28	0.215
PA	3.1	2.6	2	✓	✓	✓	✓	1.68	3.30	[1, 5]	0.30	0.233
PA	3.5	2.6	0	×	✓	×	×	1.74	1.93	[0, 3]	0.26	0.205
PA	3.5	2.6	1	×	✓	×	×	1.74	1.99	[1, 3]	0.25	0.198
PA	3.5	2.6	2	×	✓	×	×	1.74	2.21	[1, 4]	0.24	0.185

Notes. ★ marks the baseline branch. A × in columns A–C marks a subgroup whose mass-function fit failed the residual-fit criterion at that family, R_V and α ; such branches are retained as explicit sensitivity cases and are never deleted, but they are not validated fits.

Appendix B: Membership validation and the distance-selection limitation

Automatic versus manual recovery. The recall benchmark against [Berlanas et al. \(2019\)](#) decomposes as follows. Their published list contains 229 members. Of these, 168 fall inside our quality-analyzable sample, and the automatic full-covariance membership posterior recovers 128 of them, or 76.2%. A further 61 are retained as manual spectroscopic quality exceptions — stars excluded by the astrometric quality criteria, chiefly on RUWE or absence from the query, but carrying independent spectroscopic evidence of membership. Total recall is thus 189 of 229, or 82.5%. The pre-specified acceptance criterion was set on the total, and the total meets it; the automatic fraction is the number to quote when the question is how well the classifier performs unaided. The adopted footprint is itself taken from [Berlanas et al. \(2019\)](#), so the spatial component of this benchmark is not independent of it either.

Why the catalogue boundary is 0.5. For a calibrated two-class posterior, $P > 0.5$ is the maximum-posterior decision rule under equal costs for admitting a field star and rejecting a member. It is therefore statistically transparent, but it is not a field-wide astrophysical standard: different catalogues should choose different boundaries when their science goals assign different costs to completeness and contamination. As a descriptive revision check, thresholds of 0.4, 0.5 and 0.6 retain 1,441, 1,331 and 1,186 clean automatic members, respectively. External-list recovery changes only from 77.4% to 75.6% across the full interval, while the mean control-to-target yield decreases from 5.4% to 2.8%. The adopted boundary is thus a reproducible central handoff, and the reported trend makes the alternative purity-completeness trade-off explicit; it is not justified by an appeal to convention.

Control fields and prior calibration. Three control fields at matched Galactic latitude were processed through the identical selection, clustering and membership model, and their aggregate yield above $P > 0.5$ is 3.7% of the association field's after area normalization. Because the same control fields were used to calibrate the mixture prior, this is an internal consistency check of high value and not a fully external false-positive rate, and Sect. 4.1 says so where it quotes the number.

Subgroup seed stability. Gaussian mixtures were fitted for $k = 2 \dots 8$ over 50 deterministic seeds and accepted on the adjusted Rand index between seeds rather than on an information criterion. Only $k = 3$ is stable. Per-star label confusion over those seeds is below a fraction of a per cent, and a refit of the entire branch grid under stable labels moves the worst baseline residual by less than the Monte-Carlo floor, which excludes mislabelling as a source of the mass-function structure.

Why the distance test is conditional. The one- versus two-component distance comparison was run on the selected members and prefers one component. The logical structure is the issue, not the statistics: parallax entered the membership mixture as a feature, so a genuine foreground group at a distinct distance would have received low membership probability and been removed *before* the distance test could see it. The test therefore constrains the depth of the sample it was given, which is what we claim, and cannot address the existence of the foreground popu-

lation, which we do not claim. Breaking the circularity requires re-deriving membership without parallax and then re-running the entire downstream chain on the resulting sample; at DR3 parallax precision, and with an intrinsic depth of order 45 pc at 1.62 kpc, that rerun would be dominated by measurement error, which is why it is deferred to DR4 rather than attempted here.

Appendix C: Extinction, age and mass-function diagnostics

Anchor distribution and kriging support. The spectroscopic anchors are distributed very unevenly between the subgroups. Cyg OB2-B contains 4 anchors inside the member sample against 59 for A and 42 for C, and its eighth-nearest anchor lies 0°374 away against 0°089 and 0°139 (Fig. 3, centre). The kriging weights sum to 1.000 for A and 0.992 for C but only 0.772 for B. This is an observational limitation of the DR3-era anchor set, not a modelling choice, and it is why the repaired prior moved B and left A and C unchanged: no per-subgroup decision was taken. The same fact surfaces three times — as B's extinction correction, as the last mass-function fit limitation, and as B's age railing against its prior support.

The anchor-versus-photometry offset. At matched sky position, prior-free broadband photometry prefers an A_V about 0.5 mag below the spectroscopically calibrated anchors, at high significance. The absolute extinction scale, and hence the mass scale, the normalization and N_{SN} , rests on the anchor calibration rather than on the photometry alone. This is a systematic on the *scale*, not on the relative structure: it moves all three subgroups together and does not produce the A/B/C ordering. Its sign is the reassuring one — the anchors read *higher*, the opposite of the failure mode in which a selection-biased anchor set would drag the scale down.

Excluded and railed age branches. Of the age branches attempted, a minority are excluded outright for insufficient stars or an unmeasurable indicator, and a further set rail against the edge of the age grid. Both classes are recorded with their exclusion reason rather than silently dropped. Cyg OB2-B's PMS indicator in particular remains unmeasurable, so the documented disagreement between its two age indicators is reported rather than resolved.

The residual map and the fit-quality definition. A cell passes the residual-fit criterion only if its calibration-window χ^2 probability is at least 0.01, its residual trend probability at least 0.05, and its worst absolute Pearson residual at most 3.0. Figure 6 shows all 54 cells with those verdicts. The failure counts quoted in Sect. 4.3 are those of the final chain, which is the chain every downstream number was computed on; an earlier accepted version of the normalization gives the same 40 of 54 total but a different distribution of *which* cells fail, and the final chain is authoritative here.

Appendix D: Ledger algorithm and convergence

Algorithm. For each branch: draw a paired (k , age) sample from the joint posterior; draw a star-formation duration offset within $\pm\delta/2$; compute the lowest turnoff over the birth window; draw a Poisson number of stars above it with mean k times the IMF integral; sample their masses from the truncated IMF; mark as dead

those whose mass exceeds the turnoff at their own birth time; convert each death to an epoch by inverting the same turnoff relation for the stellar lifetime; and apply the explodability filter. Because the lifetime is the inverse of the turnoff relation used for the ages, the ledger cannot disagree with the age fit about when a star of a given mass dies.

The pre-registered convergence test failed. The pre-registered convergence criterion was *relative* and it **failed**: the stored execution record carries the failed verdict, and it has not been reinterpreted. The failure was driven by the criterion's own specification rather than by numerical instability — a relative tolerance around cells whose value is zero or near zero cannot be met by any number of iterations, and the ledger contains many such cells by construction, since whole branches return exactly zero. Reported instead, and *in addition* rather than in substitution, is the absolute drift of material cells under a doubling of the iteration count, which is small. The honest summary is that the analysis is numerically stable and that the criterion as written was not a good test of that, and the record retains both statements.

The threshold scan. The direct-collapse scan reruns the same populations under a hard cutoff at each of a series of threshold masses, filtering the same draws rather than re-sampling, so the scan and the baseline share their Monte-Carlo realization.

Appendix E: External checks and the conditional cocoon calculation

Pre-registered predictions. All five pre-registered marker predictions passed and their statistics and pass conditions were fixed before the ledger existed. They are reported in Sect. 4.7 with their outcomes unchanged.

Pulsar assumptions. The pulsar argument assumes that PSR J2032+4127 is associated with Cygnus OB2, which the identification of its companion with a star in our own census supports but does not prove; that a neutron star requires a successful explosion; and that the characteristic age, widened for orbital acceleration, brackets the true age. Three interpretations remain degenerate — high-mass explodability, an older population, or binary stripping of a lower-mass progenitor — and all three give a non-zero budget.

The three cocoon terms. Table A.3 states what each of C_1 , C_3 and C_4 is, how it is treated numerically, and its dominant limitation. The essential points are that only C_1 is a sampled probability; that C_3 returns unity because the ledger's mass floor lies above the adopted threshold and not because nature guarantees stripping; and that C_4 is an upper bound applied as a constant, whose sensitivity is given in Sect. 5.4 and Fig. 10.

The neighbouring-association budget, attempted and withdrawn. The original plan called for a coarse literature-based supernova budget for the neighbouring Cygnus populations. We attempted it, from the tabulation of Martin et al. (2010) after Knödlseeder et al. (2002), which gives for each of 10 populations an observed massive-star count in a stated mass interval together with a distance and an age. The method normalizes a power law to that count truncated at the turnoff, integrates above the turnoff

for the cumulative deaths, and differences over 100 kyr for the recent rate, using this project's own turnoff relation so that only the inputs differ from the measured ledger.

The pre-registered validation test required the same estimator, applied to those authors' own Cyg OB2 row, to reproduce our measured ledger to within an order of magnitude. It **failed**. At their adopted 2.5 Myr the turnoff lies above the IMF ceiling and the estimator returns identically zero against our measured 8.43 — and run at the age we measure, it overshoots by a factor 2.1–3.9. For reference, the measured ledger expects 0.80 supernovae in the last 100 kyr on the baseline branch, which is the quantity a neighbour budget would have had to be compared against. Both failures have the same cause: the estimator is dominated by the assumed age, and heterogeneous literature ages for the neighbours are precisely what no homogeneous census yet supplies. Per the pre-registration, the budget is therefore not reported as a number and the cavity-source question is scoped out of the paper's claims. The attempt, its inputs and its failed validation record are retained in the released records so that the decision can be audited rather than taken on trust.

Frozen evidence versus post-hoc forecast. The independent markers were frozen at data acquisition, before the normalization, the census and the ledger existed. The ^{60}Fe forecast of Sect. 5.8 was not: it was chosen after the ledger existed and pre-registered before it was scored. These are different kinds of evidence and the text distinguishes them wherever it uses them.

Appendix F: Provenance and reproducibility

Every cut, threshold, catalogue version and query is recorded with its date in the project's provenance log, and every validation verdict exists as a machine-readable record. Nothing is overwritten: superseded artifacts are retained with version suffixes and stay referenced, and the released input resolver refuses the superseded products by name so that a stale table cannot reach the manuscript even by accident. The manuscript's numbers are generated directly from those artifacts; no number in this paper was typed by hand, and a validator fails the build on any bare number in running text, on any macro that stops being used, and on any reference to a forbidden input.

Three caveats belong here rather than in a README. First, the final authorized input manifest matches the files on disk, but several older manifests from the early analysis stages describe superseded states and are retained as historical records rather than as current checksums; they are labelled accordingly. Second, the pre-registration for the post-hoc isotope forecast was a timestamped local specification written shortly before the scoring execution, with its prior qualitative expectations explicitly disclosed. It was *not* committed before execution and it carried no input hashes, and we describe it as what it was rather than claiming a stronger status; the revision analysis that produced the fit-quality landscape, the closure slopes, the scenario-score sensitivities and the neighbour-budget attempt does carry hashed inputs and was written before any of its results existed. Third, the released tree contains the code and the versioned intermediate tables; the bulk survey working tree is regenerated from the recorded archival queries.

Appendix G: Withdrawn and failed results

A chain that reports only its successes cannot be audited. The following are retained in the released records and must not be quoted as results.

1250 **Withdrawn intermediate results.** A runaway census computed from absolute rather than peculiar proper motions, which measured the association’s bulk drift rather than ejection and was invalidated by comparison against a canonical ejected star. A set of closure ratios whose forward integral was truncated at the same threshold its response scatters across, which absorbed most of an apparent excess. And an ^{26}Al ejecta mass whose denominator was wrong by about two orders of magnitude; the corrected comparison is quoted in flux space in Sect. 4.7, where no mass conversion of the measurement is needed, and the erroneous value
1260 is retained as withdrawn.

Failed pre-registered predictions. The ledger’s relative convergence criterion (Appendix D); one ledger prediction about the mass floor, which failed and caught a genuine over-generalization about the minimum progenitor mass on the widest star-formation-window branches; one isotope prediction that passed only vacuously and whose corrected form failed; and the neighbouring-population validation test of Appendix E. Each is recorded with its original wording and its outcome. None was reinterpreted after the result was seen.

1270 **Defects found and fixed within the revision analysis.** Two coding defects were found during the revision analysis itself, both by comparison against an independently computed quantity rather than by inspection: an inverted monotonicity assumption in the closing-slope inversion, and an inverted numerator and denominator in the star-weighted association closure. Both were fixed before any result was reported, neither affects an upstream number, and both are recorded in the released execution records.